Chapter 10
Threads and Fasteners

Topics

Exercises

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Threads & Fasteners: Topics

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10.1) Fasteners

10.2) Screw Thread Definitions

10.3) Types of Thread

10.4) Manufacturing Screw Threads

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10.6) Unified Threads

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Threads & Fasteners: Exercises

Exercise 10-1: Screw thread features

Exercise 10-2: Unified national thread note components

Exercise 10-3: Unified national thread note

Exercise 10-4: Metric thread note components

Exercise 10-5: Metric thread tables

Exercise 10-6: Fastener tables and clearance holes

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Threads and Fasteners

Summary

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Summary

- **What will we learn in Chapter 10?**
  - → How to represent fasteners and threads on an engineering drawing.
  - → How to calculate bolt and screw clearance holes.

- **Key points**
  - → Threads are represented by thread symbols, not by a realistic drawing.

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Threads and Fasteners

10.1) Fasteners

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Fasteners

- **Fasteners include:**
  - → bolts and nuts (threaded)
  - → set screws (threaded)
  - → washers
  - → keys
  - → pins



- **Fasteners are not a permanent means of assembly such as welding or adhesives.**

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Fasteners

- **Fasteners and threaded features must be specified on your engineering drawing.**
  - → Threaded features: Threads are specified in a thread note.
  - → General Fasteners: Purchasing information must be given to allow the fastener to be ordered correctly.

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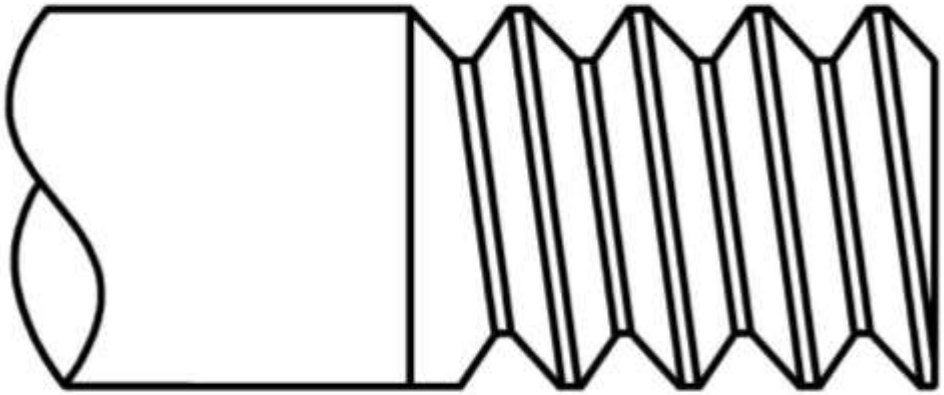
Threads and Fasteners

10.2) Screw Thread Definitions

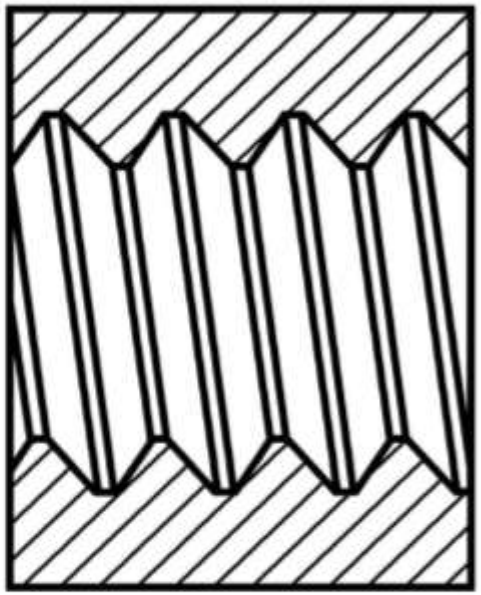
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Thread Definitions

- **Screw Thread:** A ridge of uniform section in the form of a helix.



External Threads



Internal Threads

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Thread Definitions

- **External Thread:** External threads are on the outside of a member.

  → A chamfer on the end of the screw thread makes it easier to engage the nut.



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Thread Definitions

- **External Thread:**



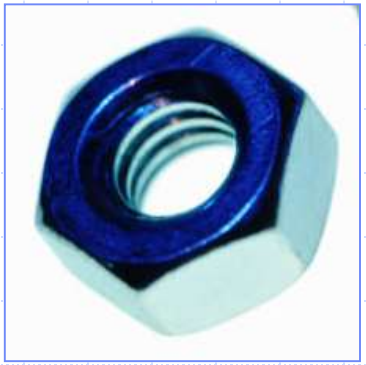
→ An external thread is cut using a die or a lathe.



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Thread Definitions

- **Internal Thread:** Internal threads are on the inside of a member.



→ An internal thread is cut using a tap.





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Thread Definitions

- **Major DIA (D):** The largest diameter (For both internal and external threads).
- **Minor DIA (d):** The smallest diameter.
- **Depth of thread:** (D-d)/2
- **Pitch DIA (d_P):** The diameter at which a line cuts the spaces and threads equally.

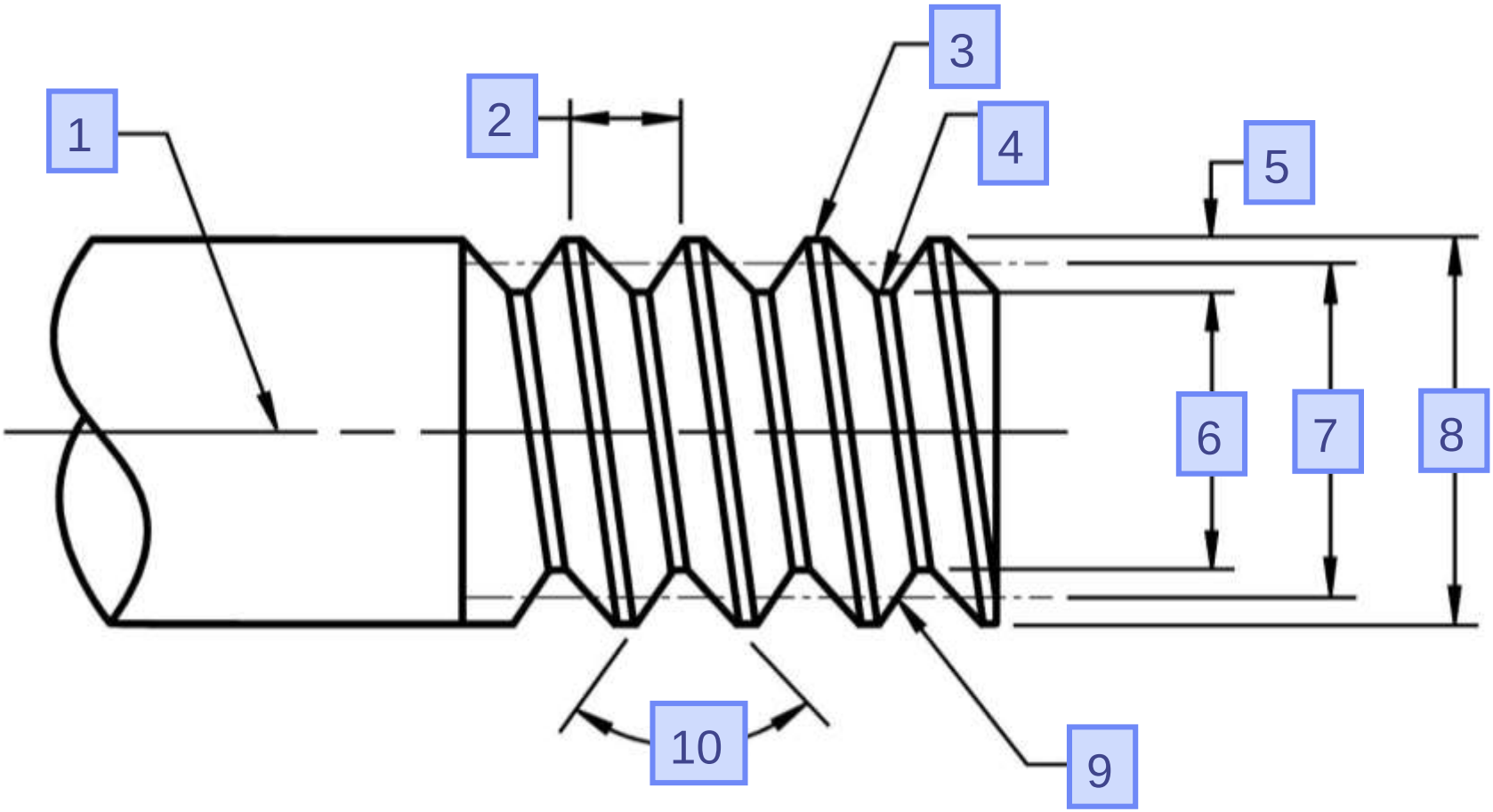
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Exercise 10-1

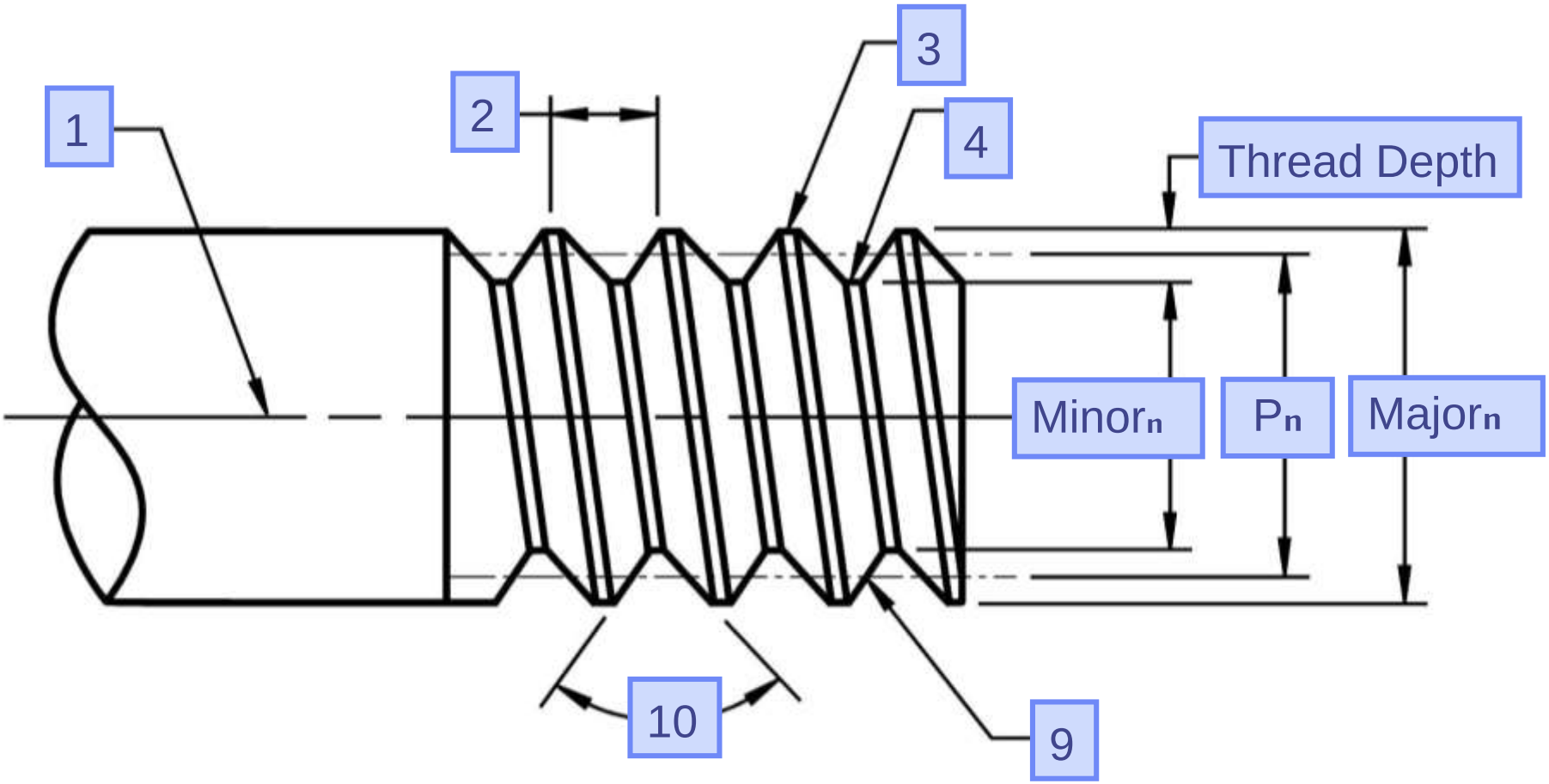
Screw thread features

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Identify the *Major, Minor & Pitch* diameters and the *Thread Depth*.



Identify the *Major, Minor & Pitch* diameters and the *Thread Depth*.

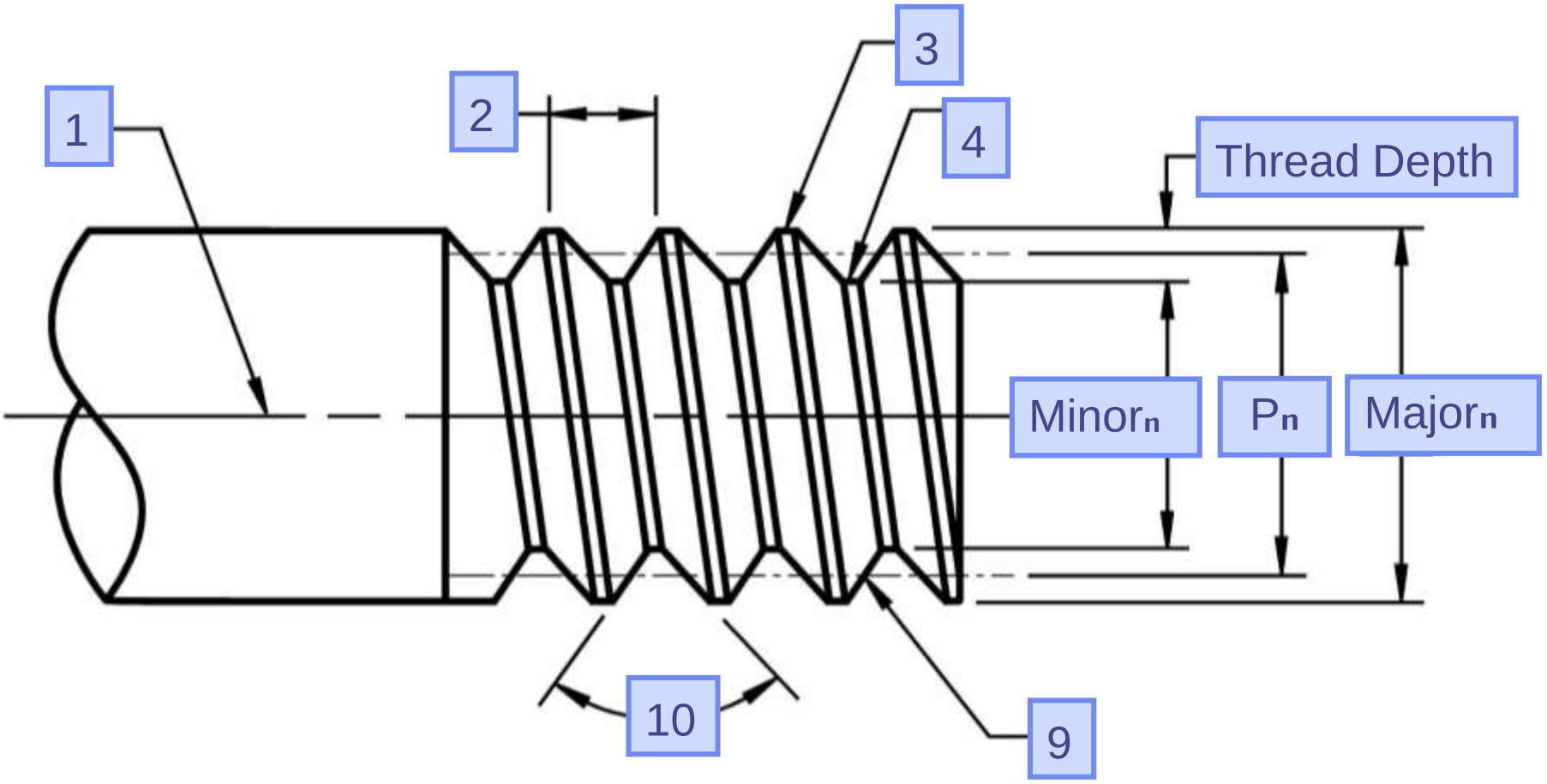


Thread Definitions

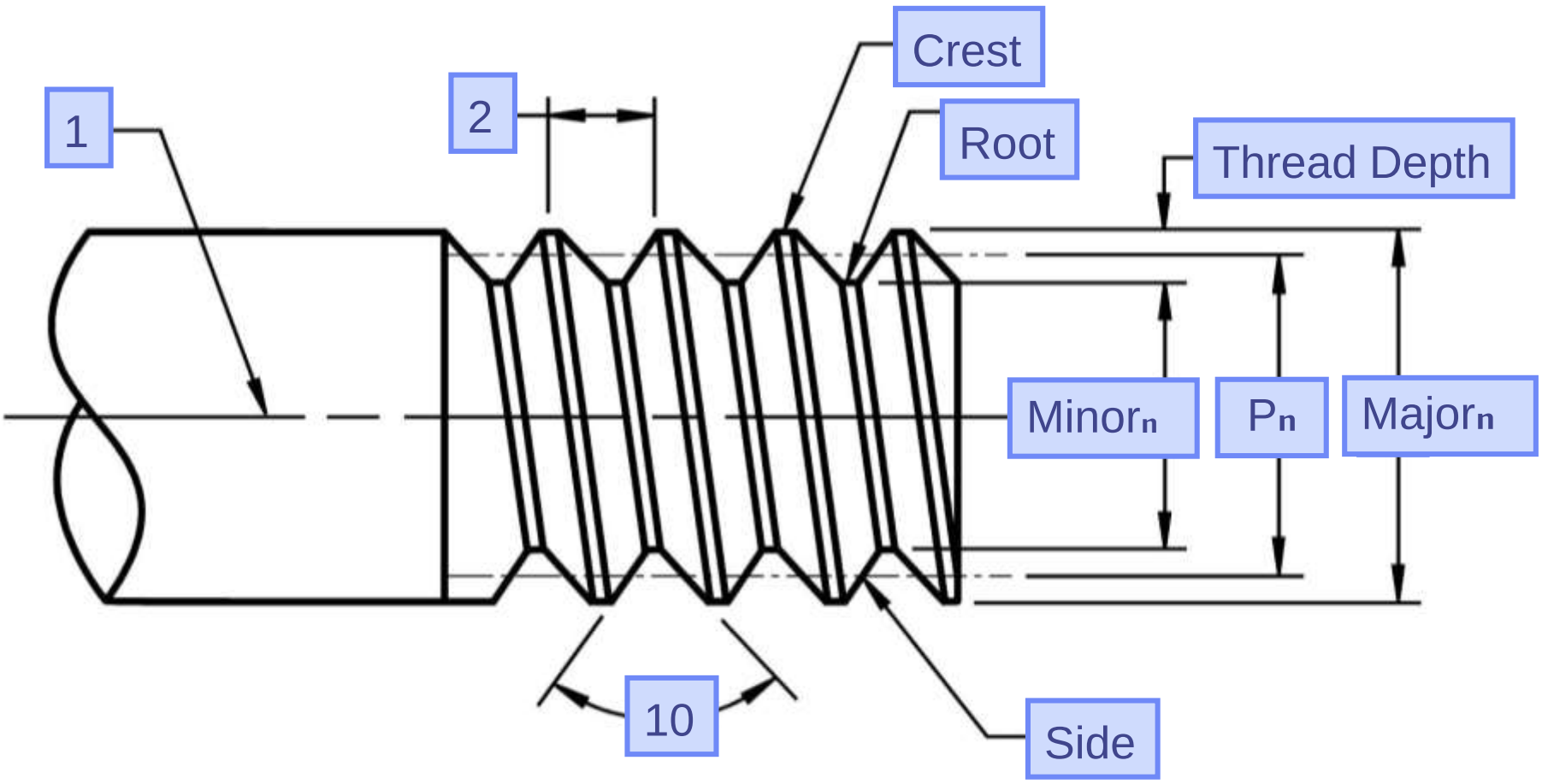
- **Crest:** The top surface.
- **Root:** The bottom Surface.
- **Side:** The surface between the crest and root.

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Identify the *Crest*, *Root* and *Side*.



Identify the *Crest*, *Root* and *Side*.



Thread Definitions

- **Pitch (P):** The distance from a point on a screw thread to a corresponding point on the next thread (in/Threads).

- **Angle of Thread (A):** The angle between the threads.

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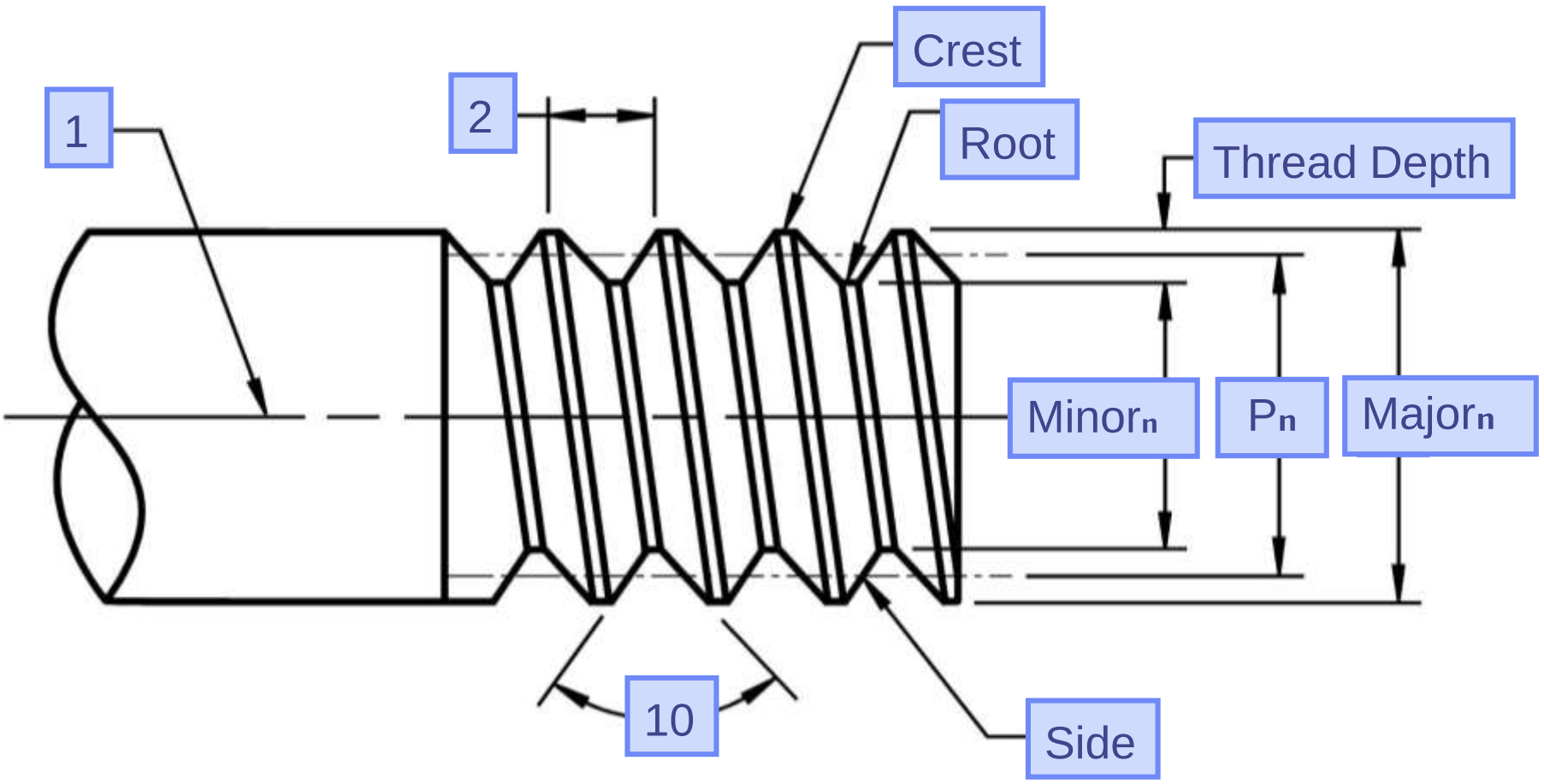
Thread Definitions

- **Screw Axis:** The longitudinal centerline.

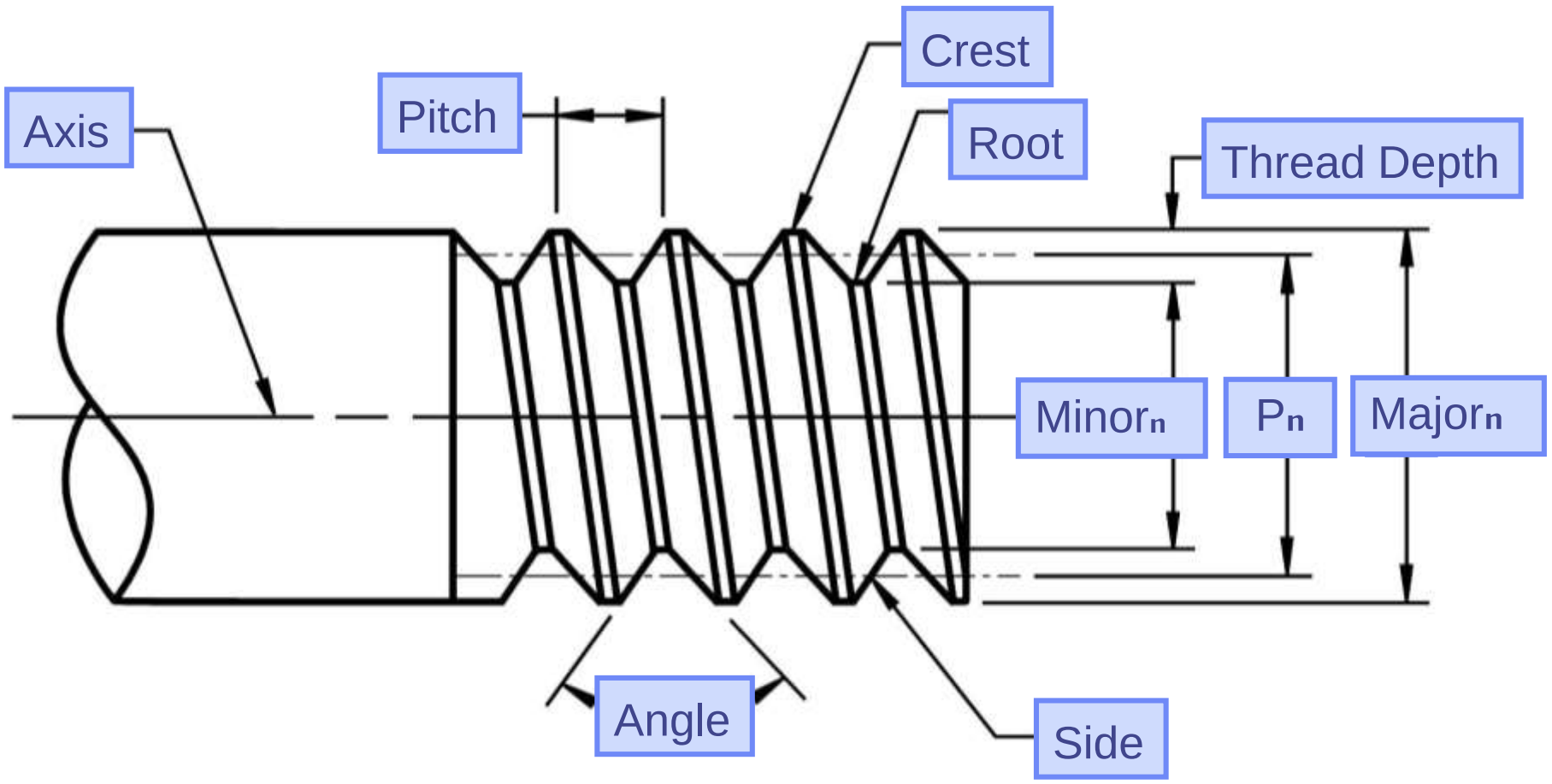
- **Lead:** The distance a screw thread advances axially in one turn.

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Identify the *Pitch*, *Screw Axis* and *Thread Angle*.



Identify the *Pitch*, *Screw Axis* and *Thread Angle*.



Thread Definitions

- **Right Handed Thread:** Advances when turned CW. (Threads are assumed RH unless specified otherwise.)

- **Left Handed Thread:** Advances when turned CCW.

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Application Question 10-1

- Name an example of a left handed thread.

  Left peddle of a bike

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Threads and Fasteners

10.3) Types of Thread

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Types of Thread

- **There are many different types of thread forms (shape) available. The most common are;**
  - → Unified
  - → Metric

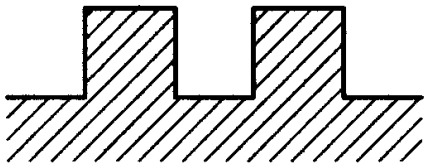
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Types of Thread

- **Thread form choice depends on;**
  - → what it will be used for
  - → length of engagement
  - → load
  - → etc…

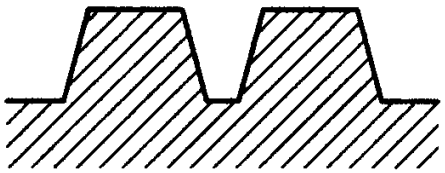
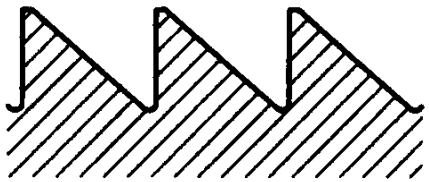
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Types of Thread (Form)

Thread Name	Figure	Uses
Unified screw thread		General use.
ISO metric screw thread		General use.
Square		Ideal thread for power transmission.

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Types of Thread (Form)

Thread Name	Figure	Uses
ACME		Stronger than square thread.
Buttress		Designed to handle heavy forces in one direction. (Truck jack)

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Threads and Fasteners

10.4) Manufacturing Screw Threads

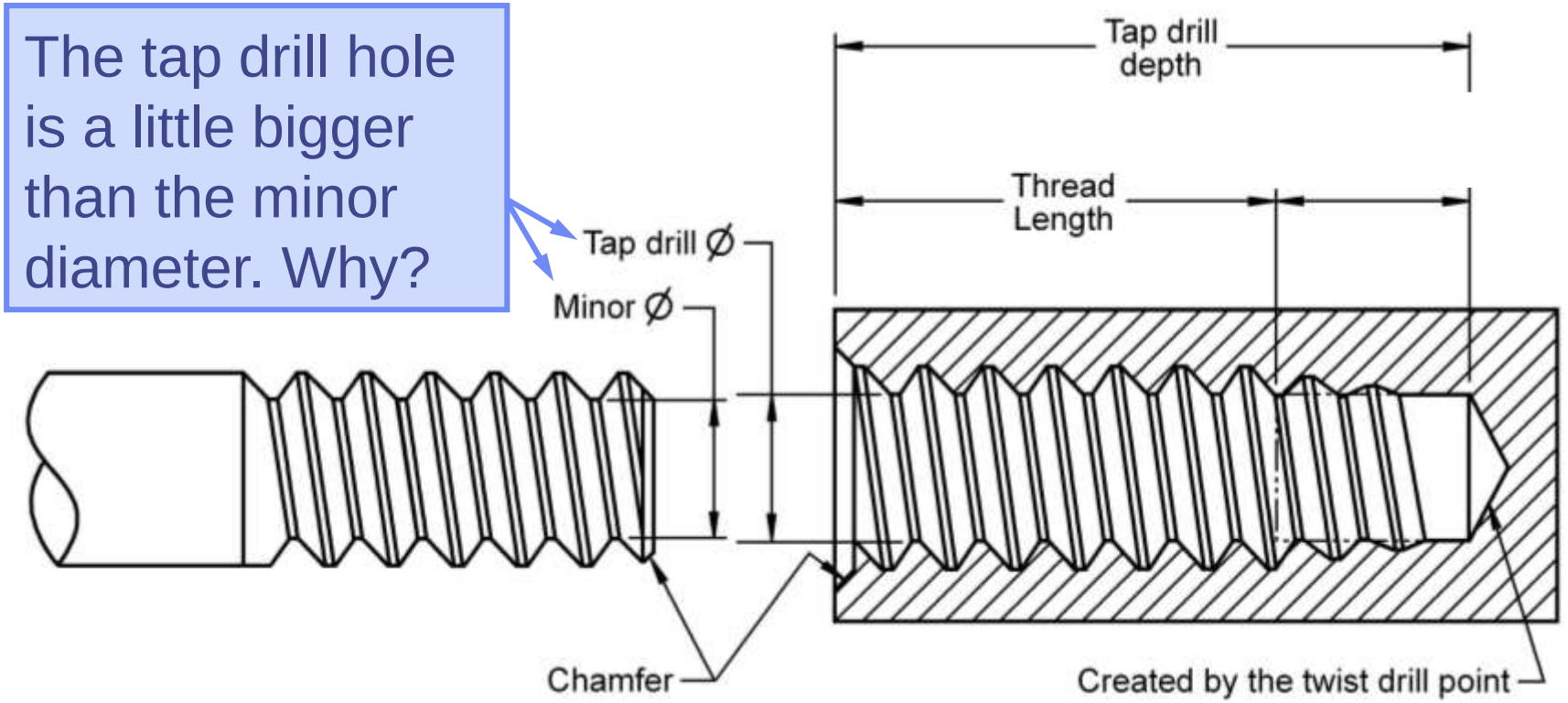
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Manufacturing Threads

➢ Internal Threads

→ First a tap drill hole is cut with a twist drill.

The tap drill hole is a little bigger than the minor diameter. Why?

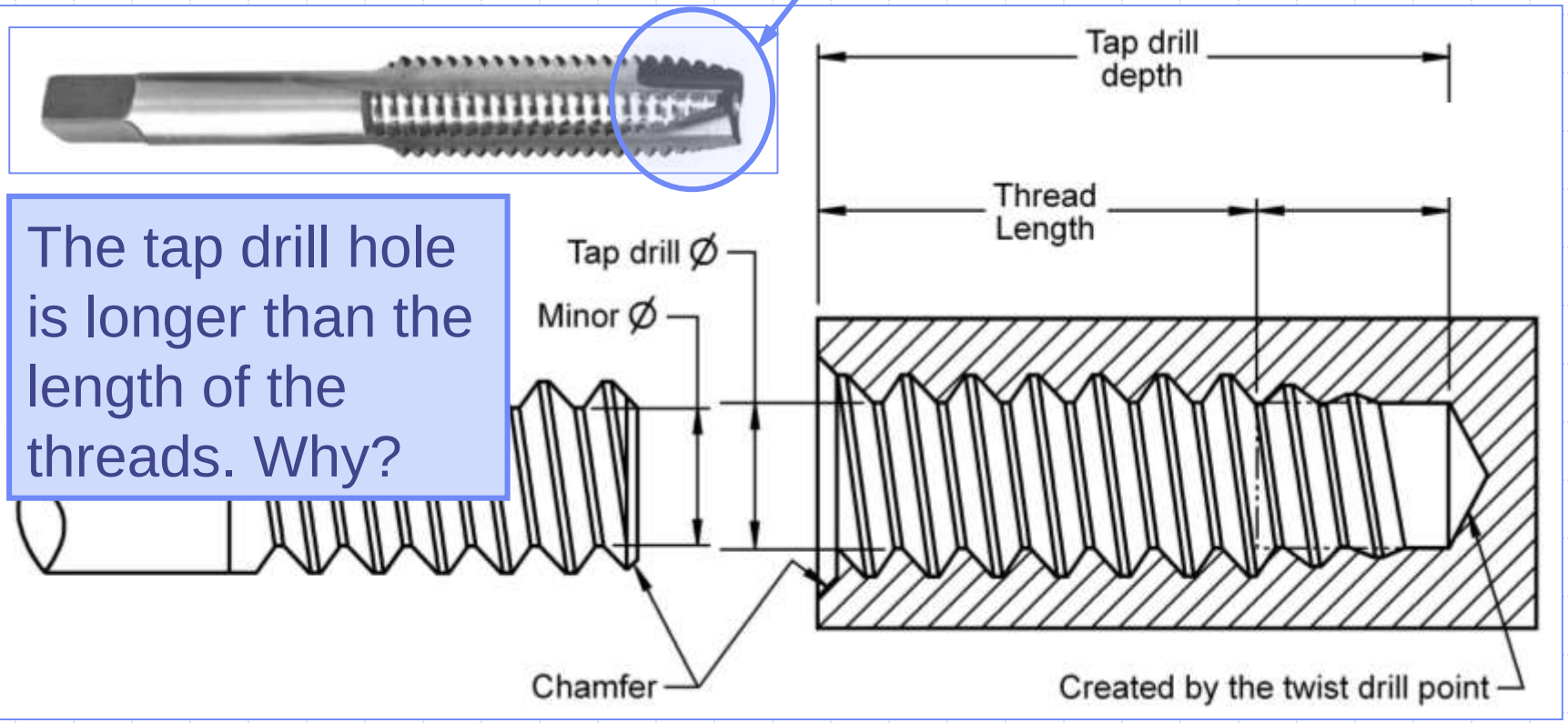


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Manufacturing Threads

- **Internal Threads**

  → Then the threads a [Incomplete threads] a tap.

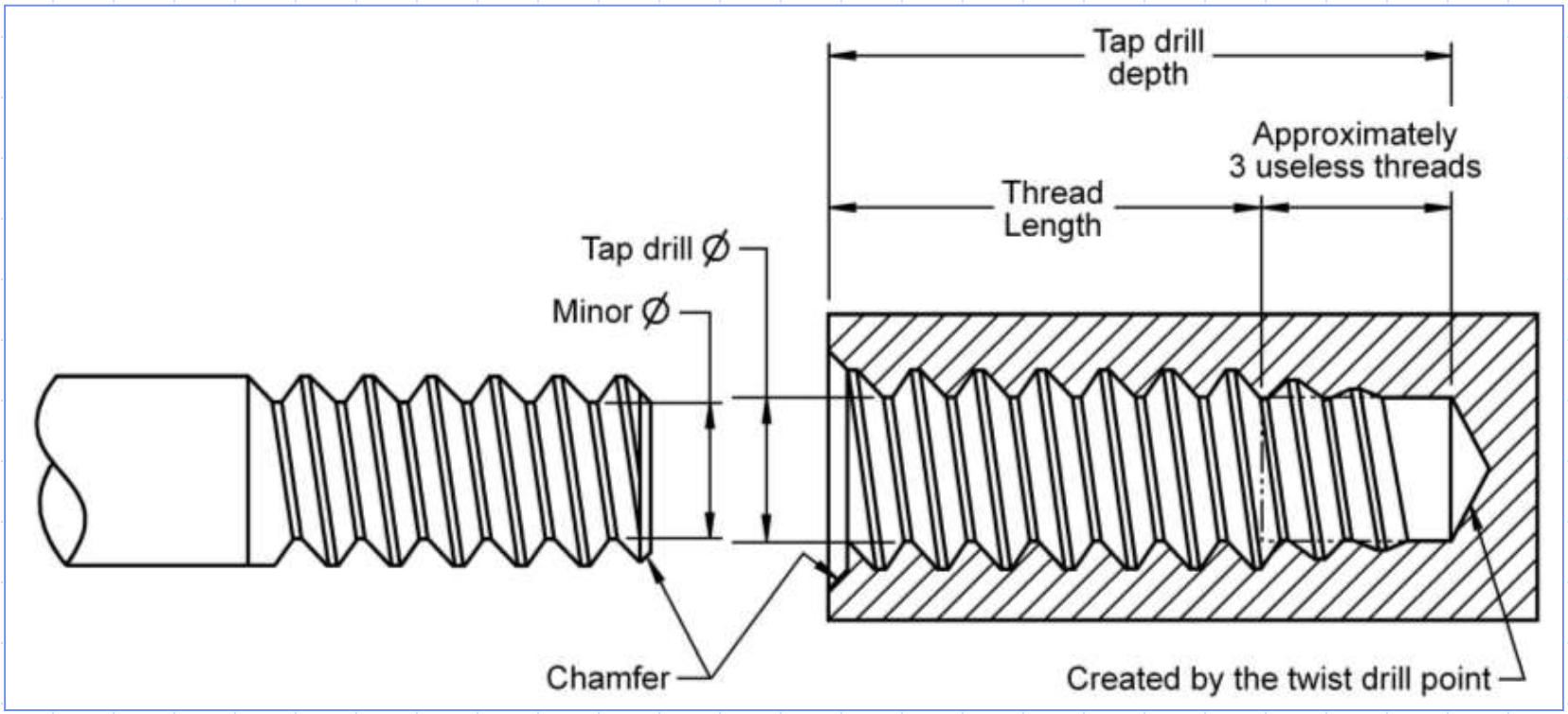


The tap drill hole is longer than the length of the threads. Why?

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Manufacturing Threads

- **Internal Threads**
  - →Chamfers are sometimes cut to allow for easy engagement.

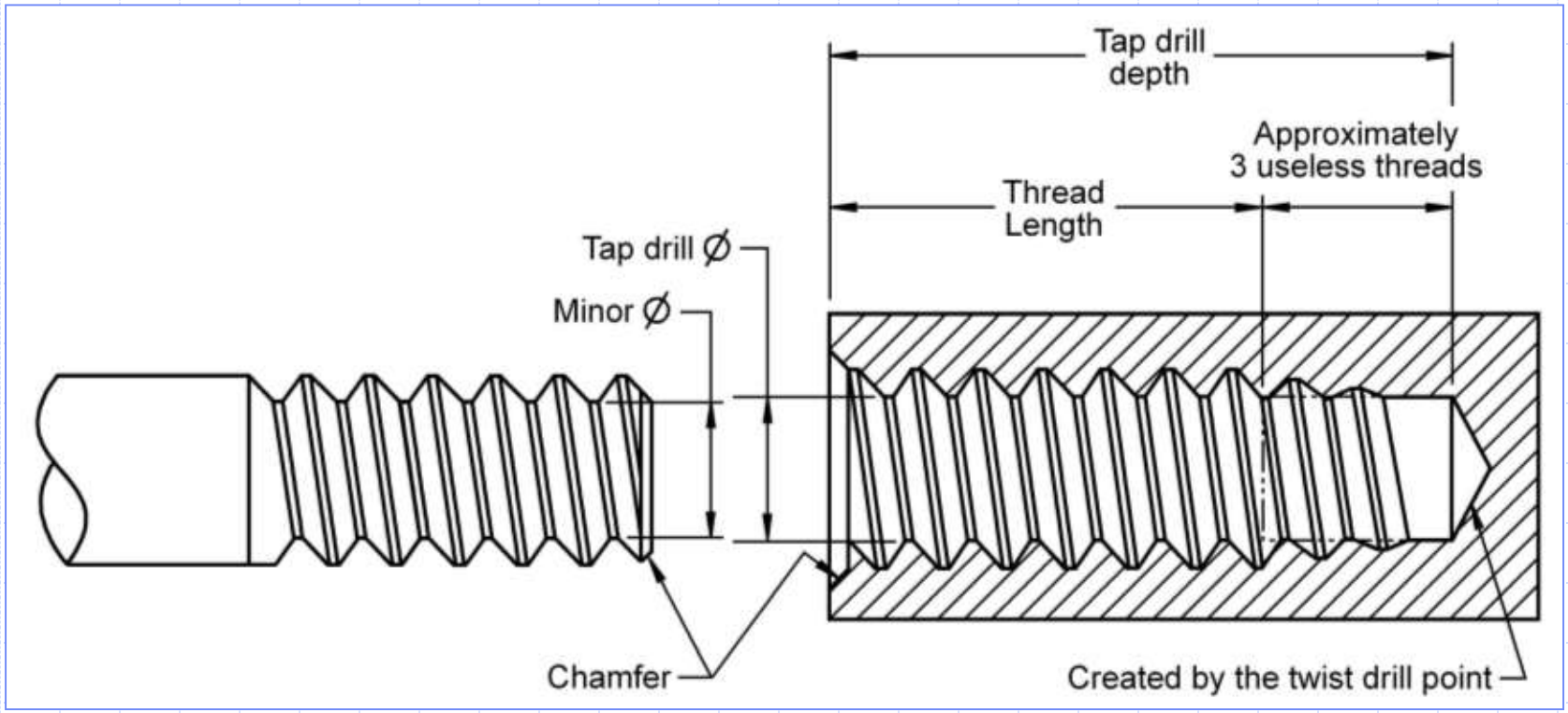


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Manufacturing Threads

External Threads

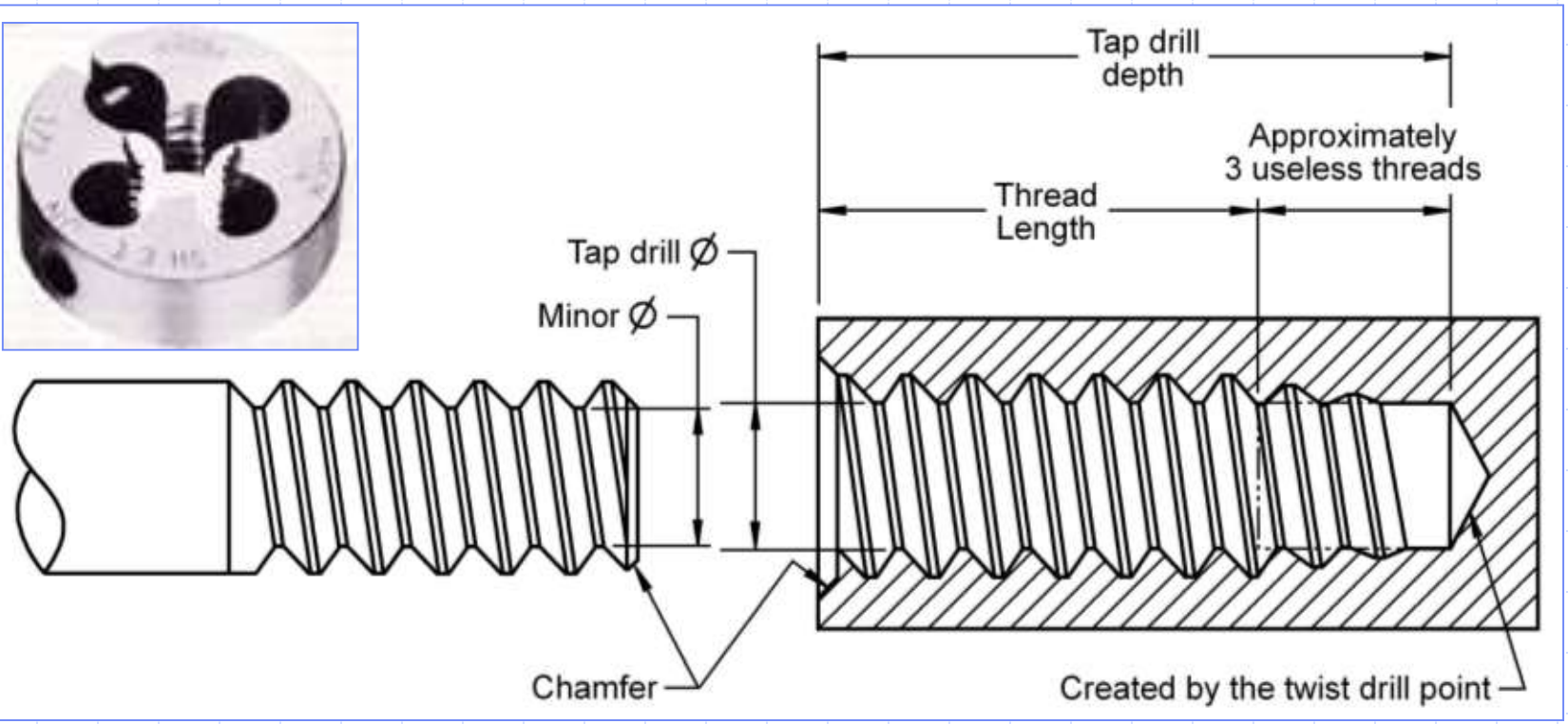
→You start with a shaft the same size as the major diameter.



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Manufacturing Threads

- **External Threads**
  - → The threads are then cut using a die or on a lathe.



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Threads and Fasteners

10.5) Drawing Screw Threads

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Drawing Screw Threads

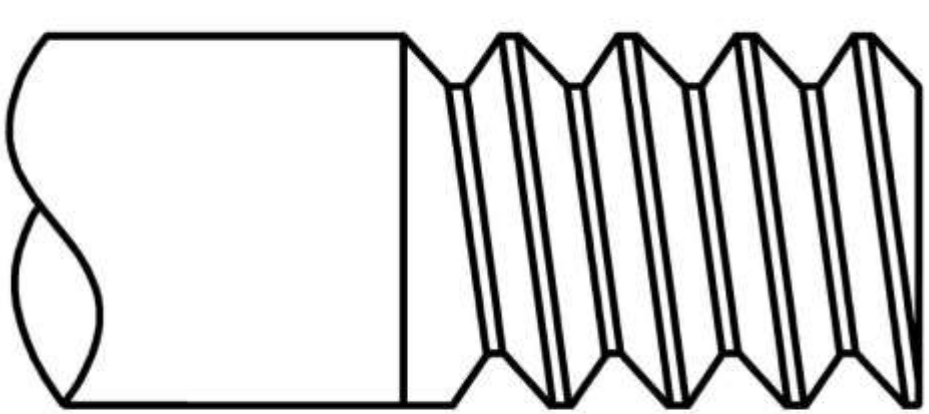
- **There are three methods of representing screw threads on a drawing.**
  - → Detailed
  - → Schematic
  - → Simplified

- Screw thread representation present in this chapter is in accordance with the **ASME Y14.6-2001** standard.

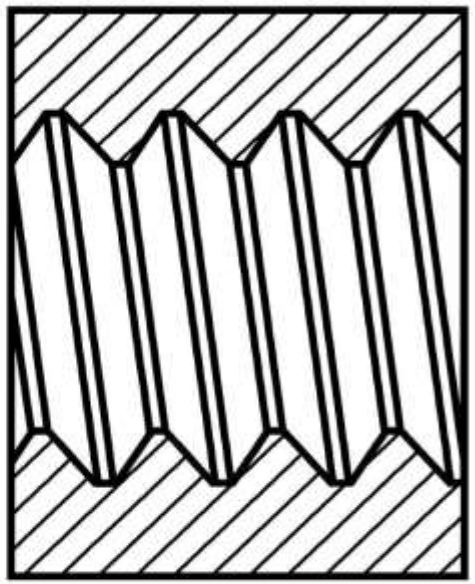
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Detailed Representation

- A ***detailed*** representation is a close approximation of the appearance of an actual screw thread.



External Threads



Internal Threads

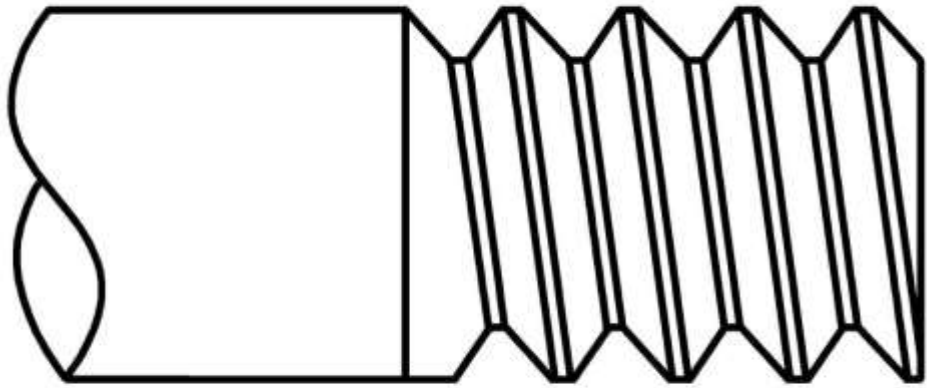
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Detailed Representation

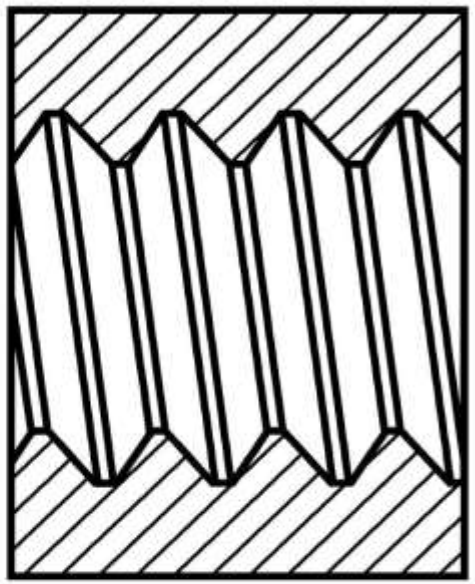
- Pros and Cons?

  Pro: Looks good and clearly represents a thread.

  Con: Takes a long time to draw.



External Threads

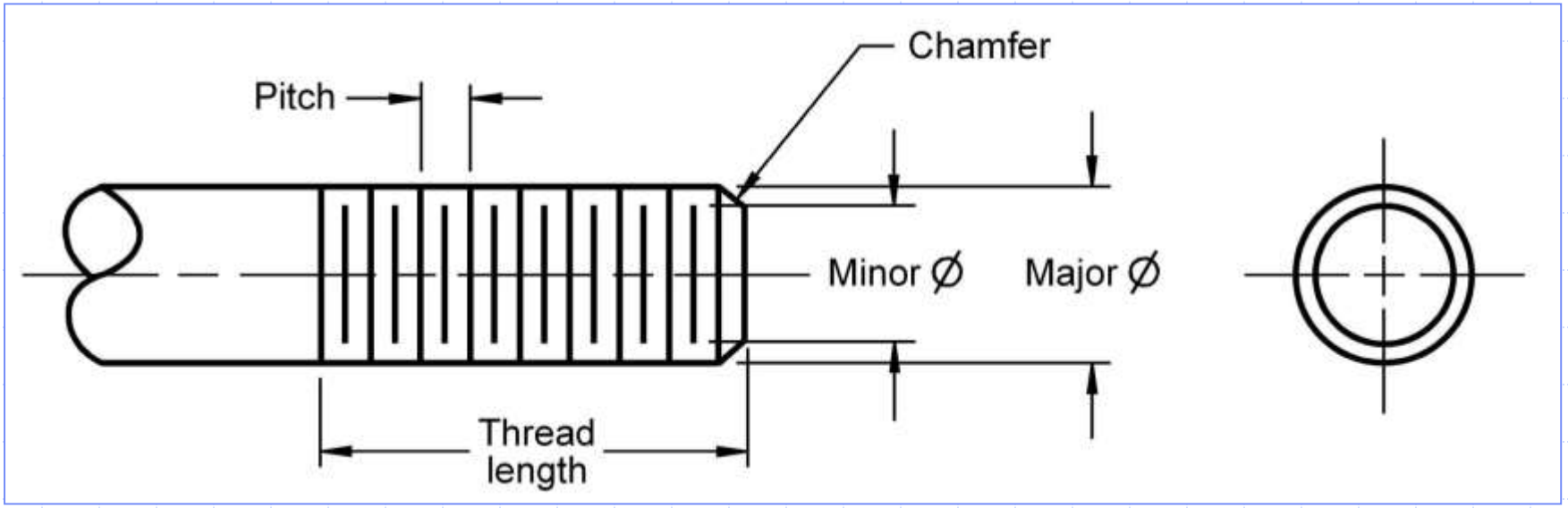


Internal Threads

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Schematic Representation

- The ***schematic*** representation uses staggered lines to represent the thread roots and crests.



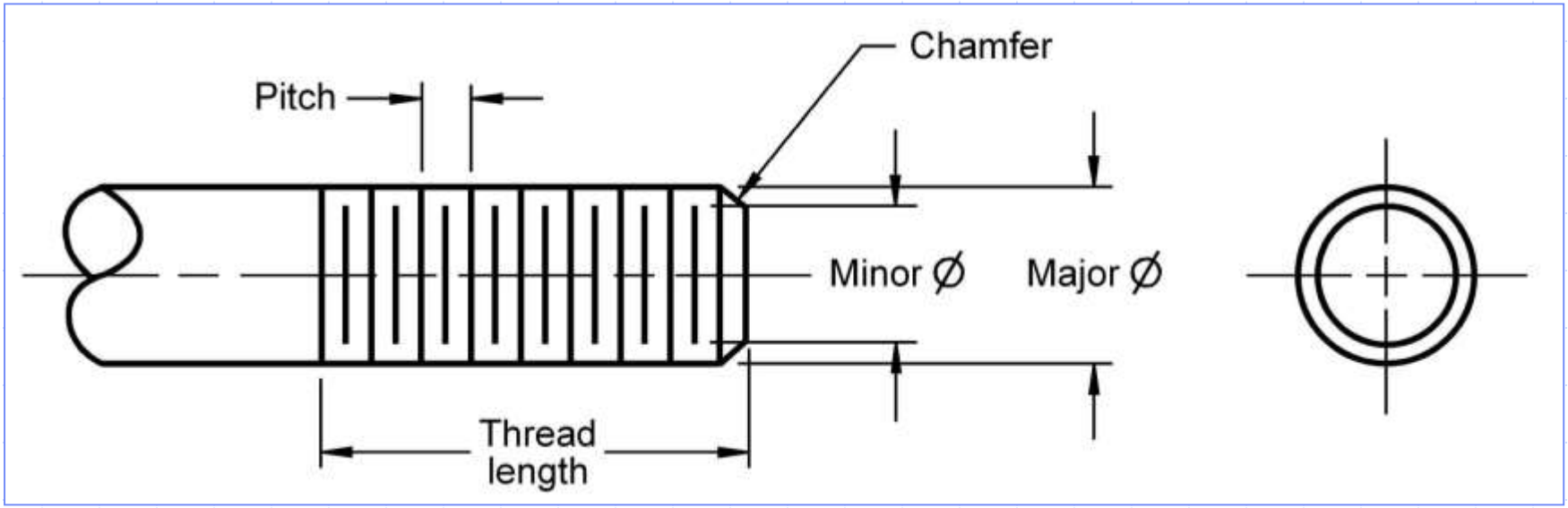
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Schematic Representation

- Pros and Cons?

Pro: Nearly as effective as the detailed representation and easier to draw.

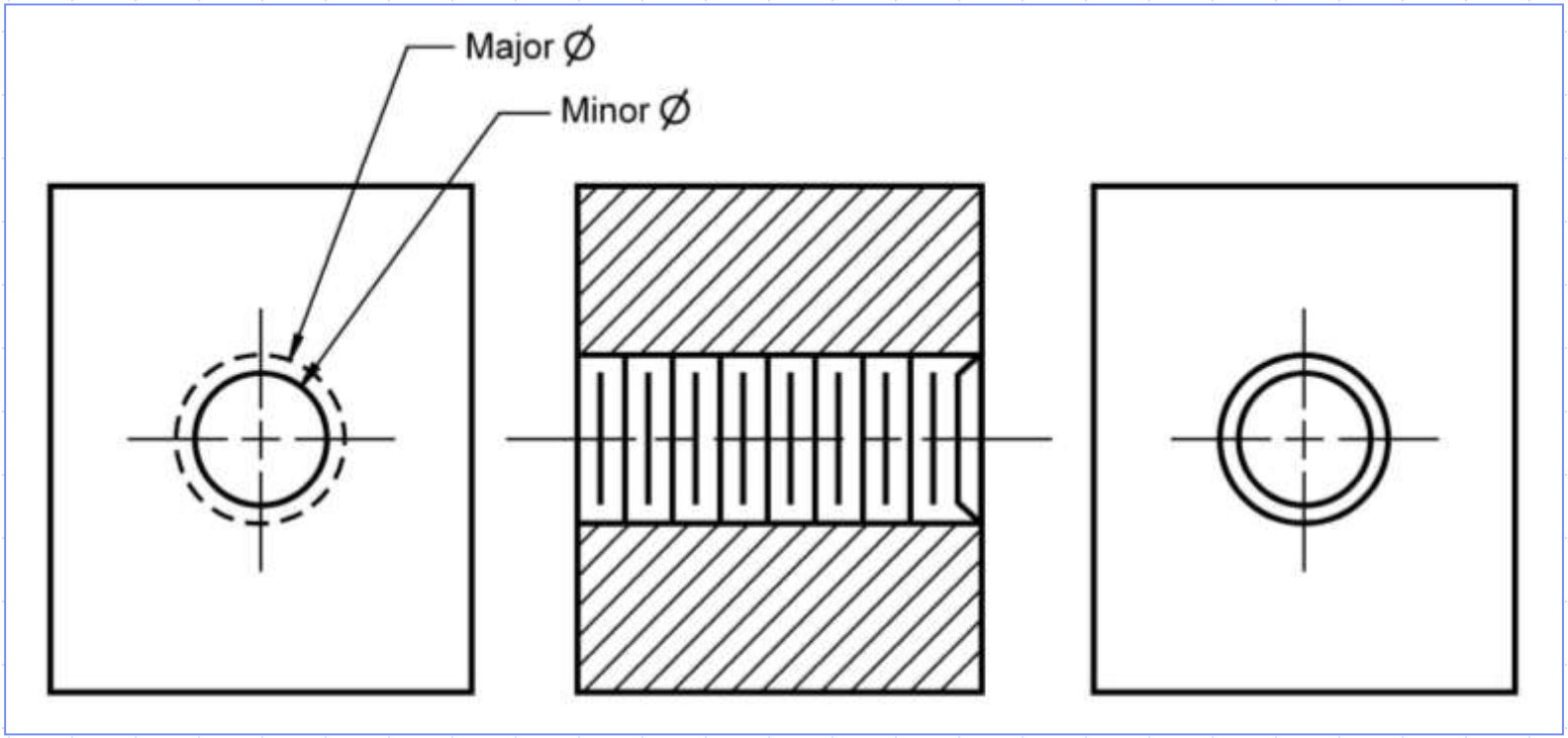
Con: Still takes some time to draw.



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Schematic Representation

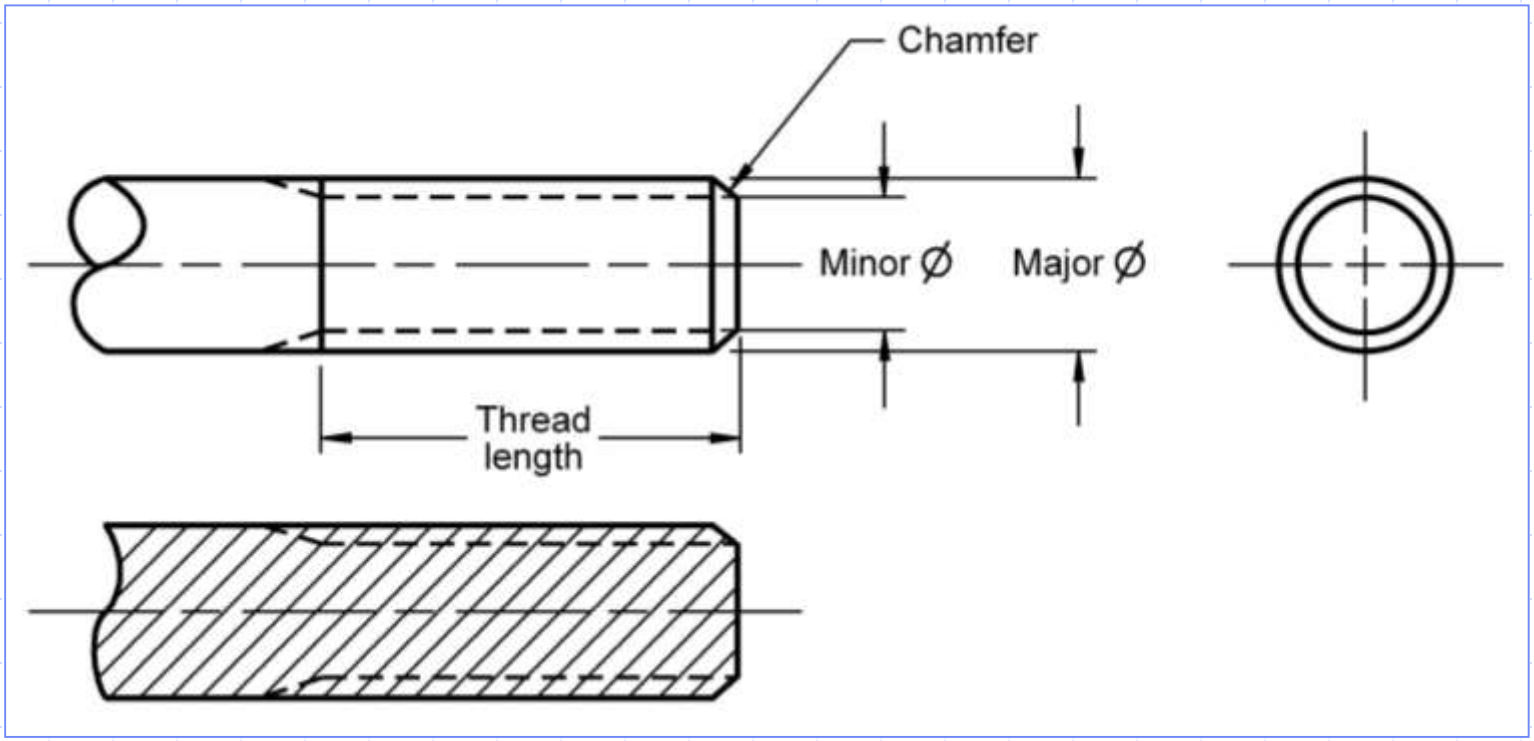
- **Rules of use for *Schematic* threads**
  - →Should not be used for hidden internal threads or sections of external threads.



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Simplified Representation

- The ***simplified*** representation uses visible and hidden lines to represent the major and minor diameters.



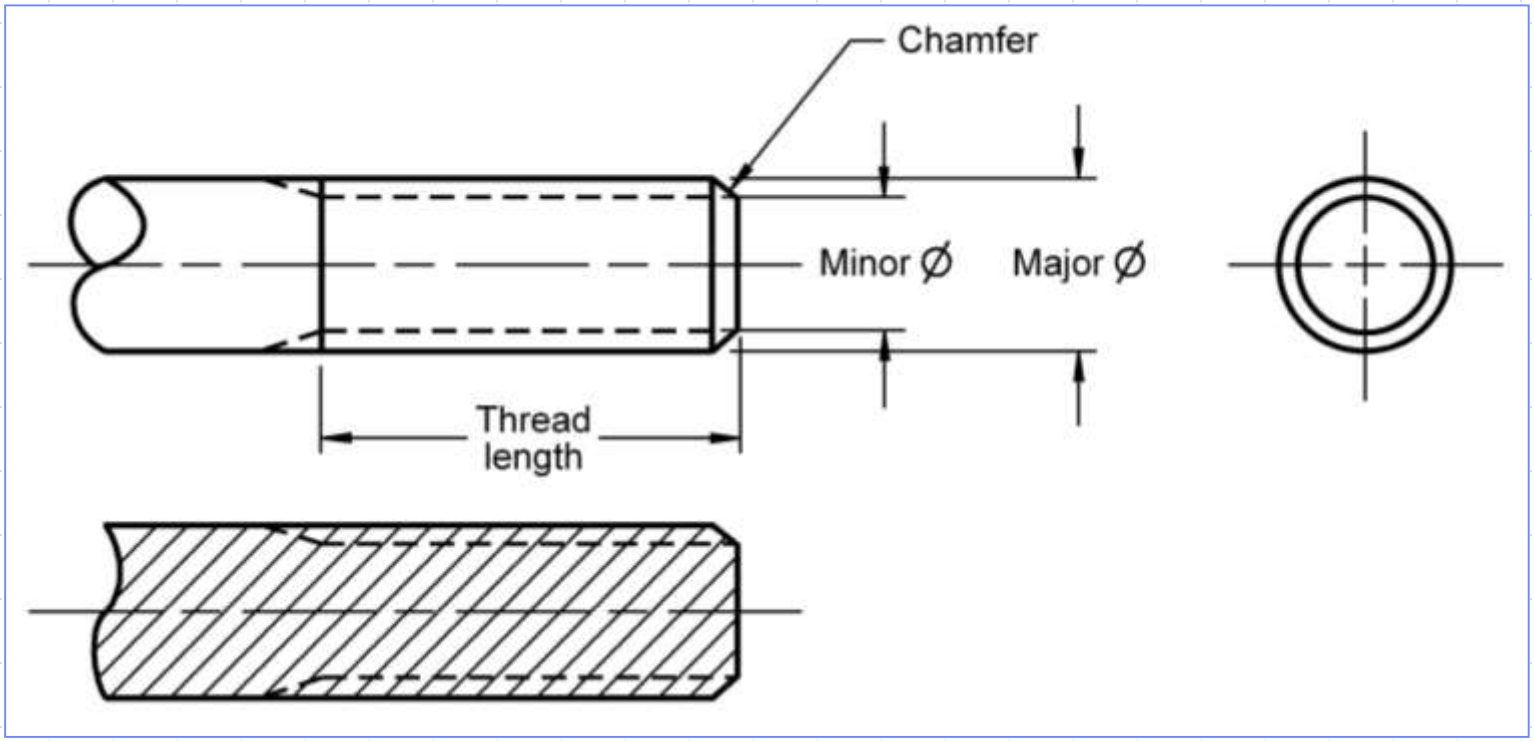
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Simplified Representation

- Pros and Cons?

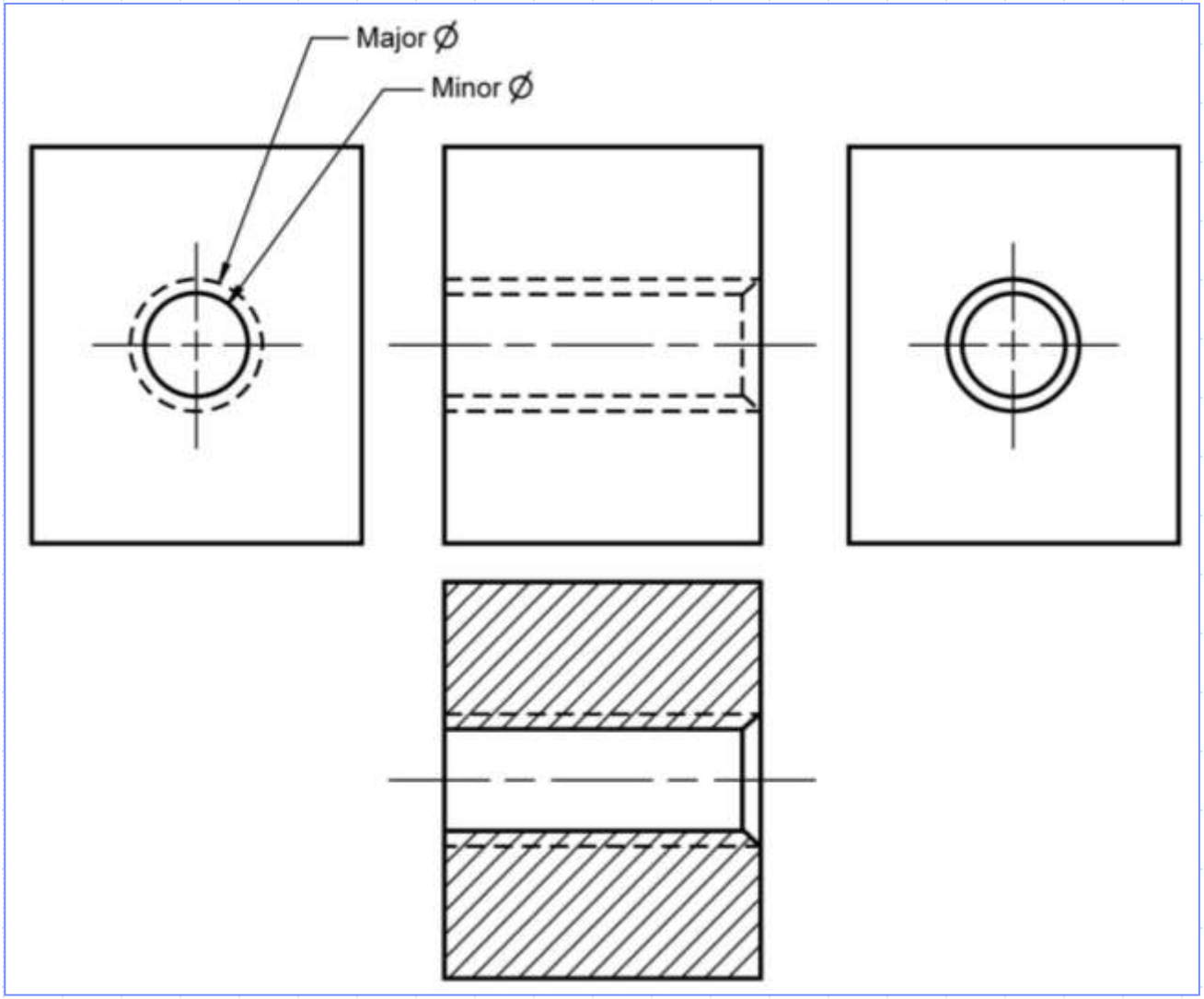
  Pro: Simple and fast to draw.

  Con: Doesn't look like a thread.



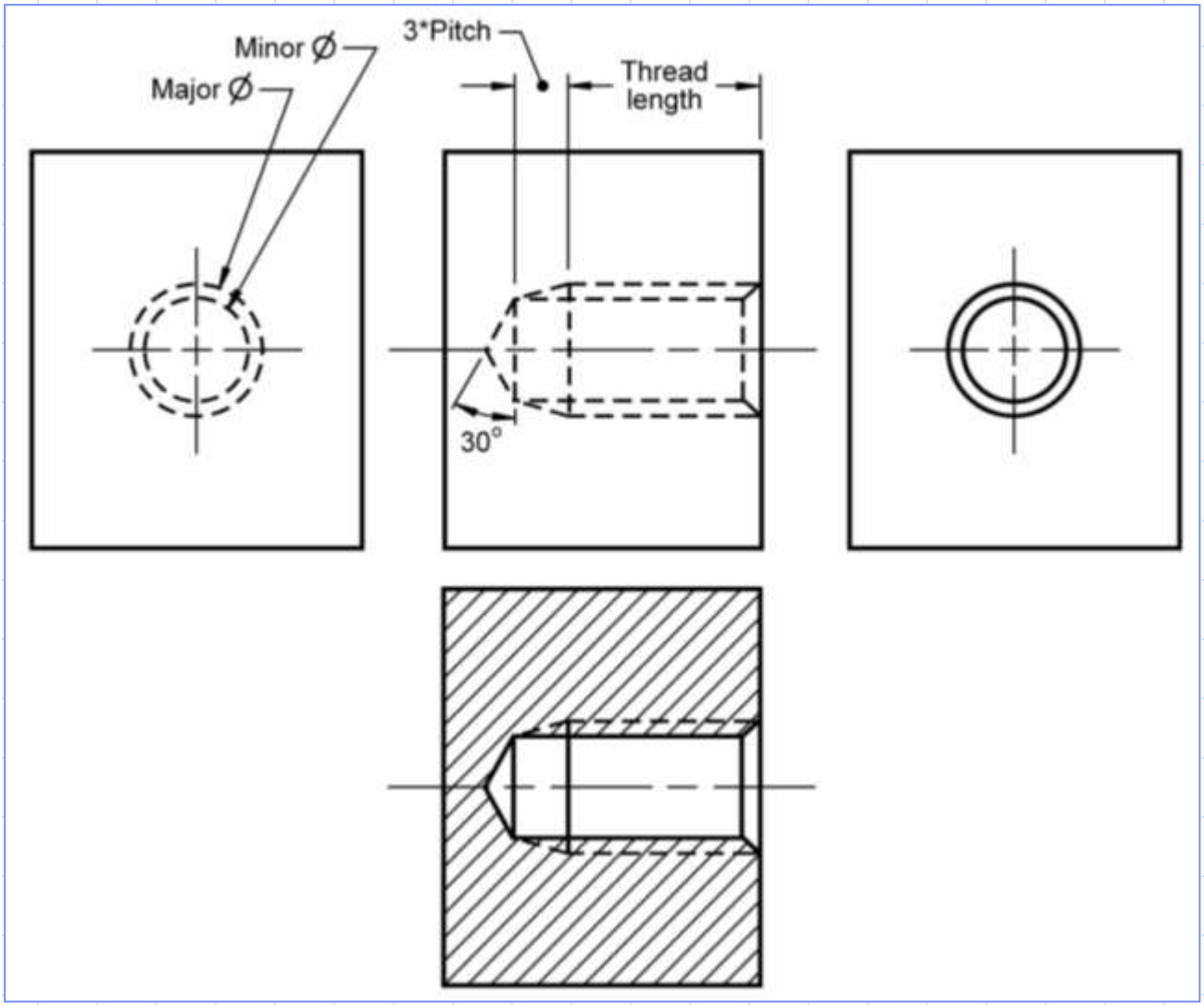
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Simplified Internal Threads



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Simplified Internal Threads



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Drawing Screw Threads

- Thread tables in the appendix can be used to look up value for the;
  - → Pitch
  - → Minor diameter
  - → Tap drill diameter

- If screw thread tables are not available, the minor diameter can be approximated as 75% of the major diameter.

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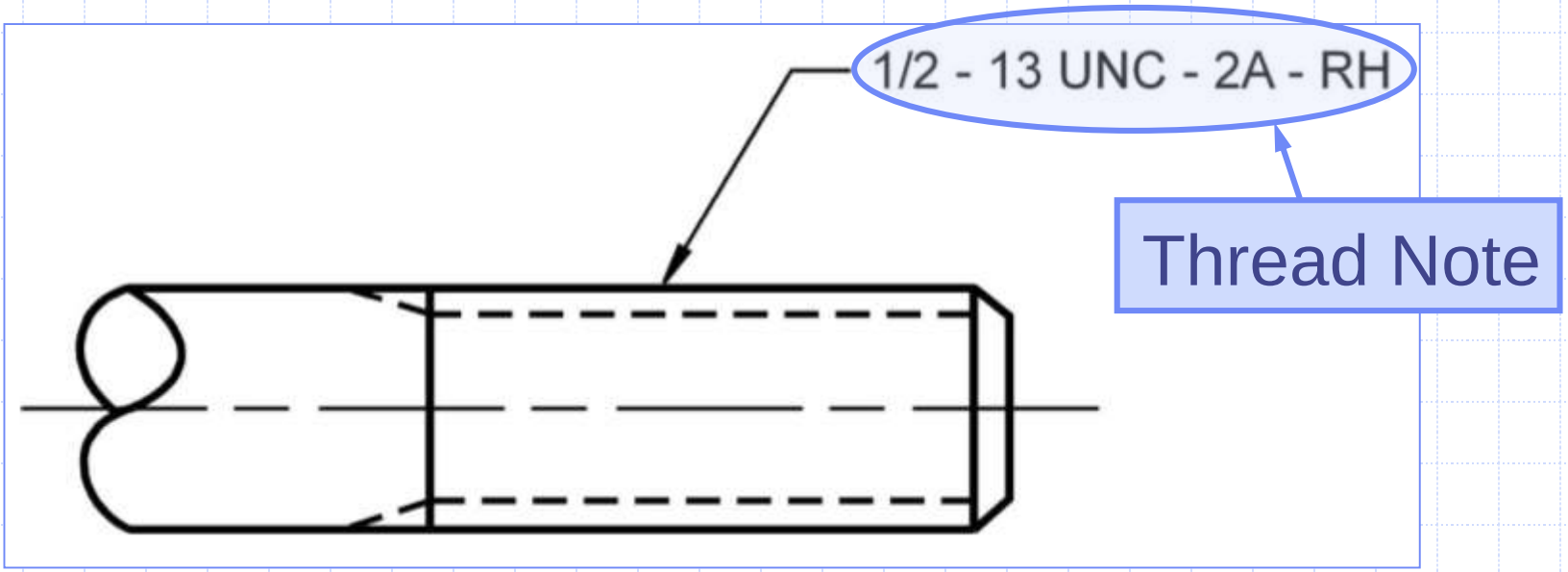
Threads and Fasteners

10.6) Unified Threads

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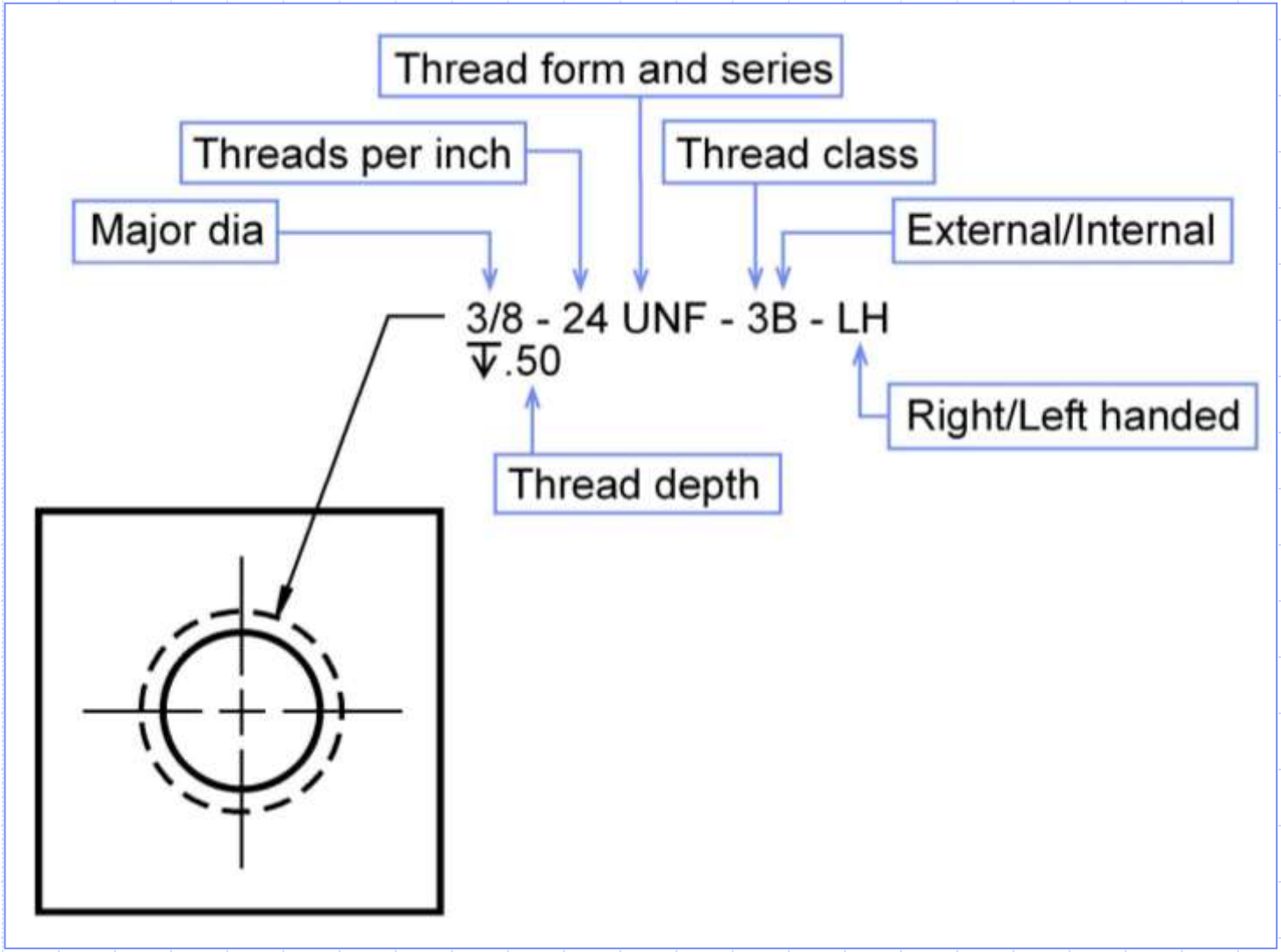
Unified Threads (inch)

- After drawing a thread, we need to identify the size and thread form in a thread note.



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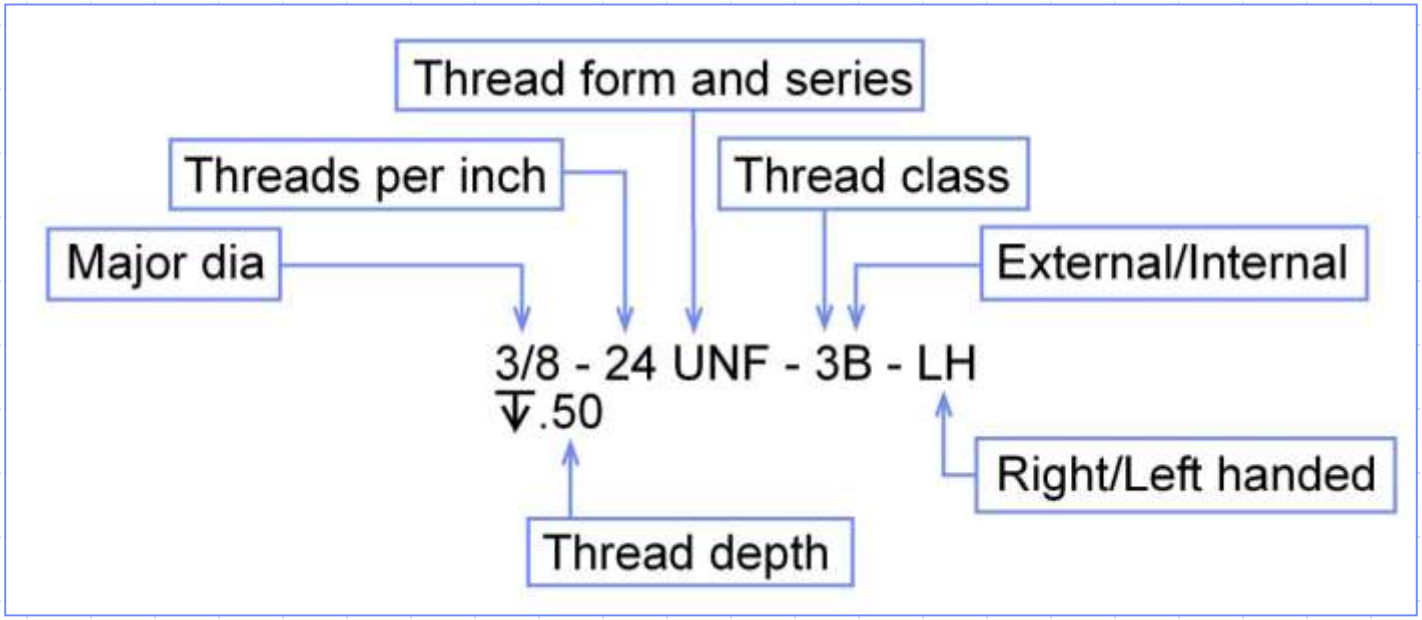
Unified Thread Note Components



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Unified Threads (inch)

- **Major Diameter:** The largest diameter.
- **Threads per inch:** Number of threads per inch for a particular diameter.
  → Equal to one over the pitch (1/P).



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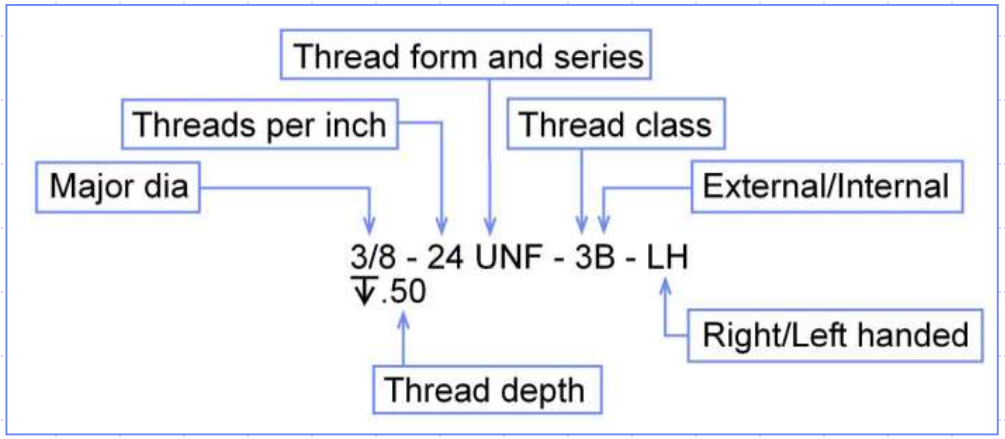
Unified Threads (inch)

- **<u>Thread Form and Series:</u>** The shape of the thread cut.
  - → **UNC** = *Unified National coarse*.
    - For general use.
  - → **UNF** = *Unified National fine*.
    - Used when high degree of tightness is required.
  - → **UNEF** = *Unified National extra fine*.
    - Used when length of engagement is limited (Example: Sheet metal).

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Unified Threads (inch)

- **Thread Class:** Closeness of fit between the two mating threaded parts.
  - → **1** = Generous tolerance. For rapid assembly and disassembly.
  - → **2** = Normal production
  - → **3** = High accuracy



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Unified Threads (inch)

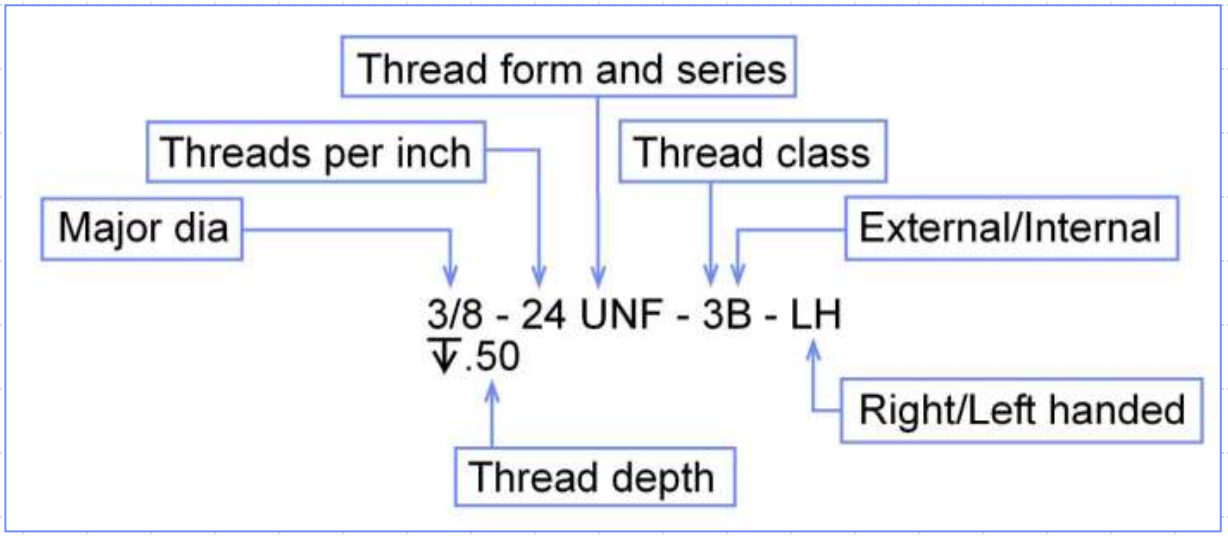
- **External or Internal Threads**
  - → **A** = External threads
  - → **B** = Internal threads
- **Right handed or left handed thread**
  - → **RH** = Right handed (right handed threads are assumed if not stated.)
  - → **LH** = Left handed

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Unified Threads (inch)

- **<u>Depth of thread:</u>** The thread depth is given at the end of the thread note and indicates the thread depth for **internal threads**

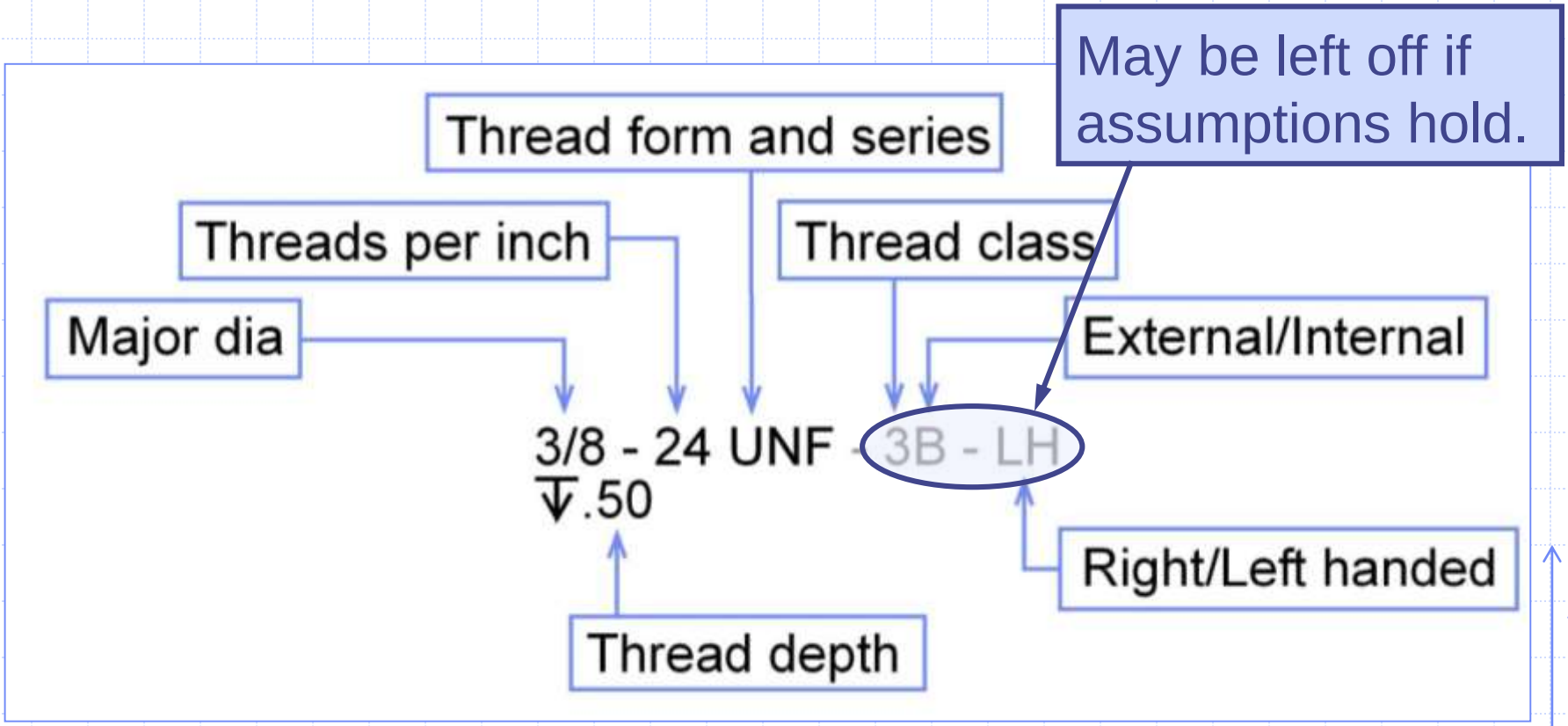
  → This is not the tap drill depth.



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Unified Threads (inch)

- Thread class is assumed to be **2**.
- Threads are assumed to be **RH**.



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Exercise 10-2

Unified National thread note components

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Exercise 10-2

- Identify the different components of the following Unified National thread note.
- **1/4 – 20 UNC – 2A – RH**

1/4	.25 inch Major DIA
20	20 threads per inch (P = 1/20 = .05)
UNC	Thread form & series – UN Coarse
2	Thread Class – Normal Production
A	External Threads
RH	Right Handed Threads

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Unified National Thread Tables

- **Standard screw thread tables are available in order to look up the:**
  - → Major diameter
  - → Threads per inch
  - → Minor diameter or Tap drill size.

- Thread tables are located in Appendix B.

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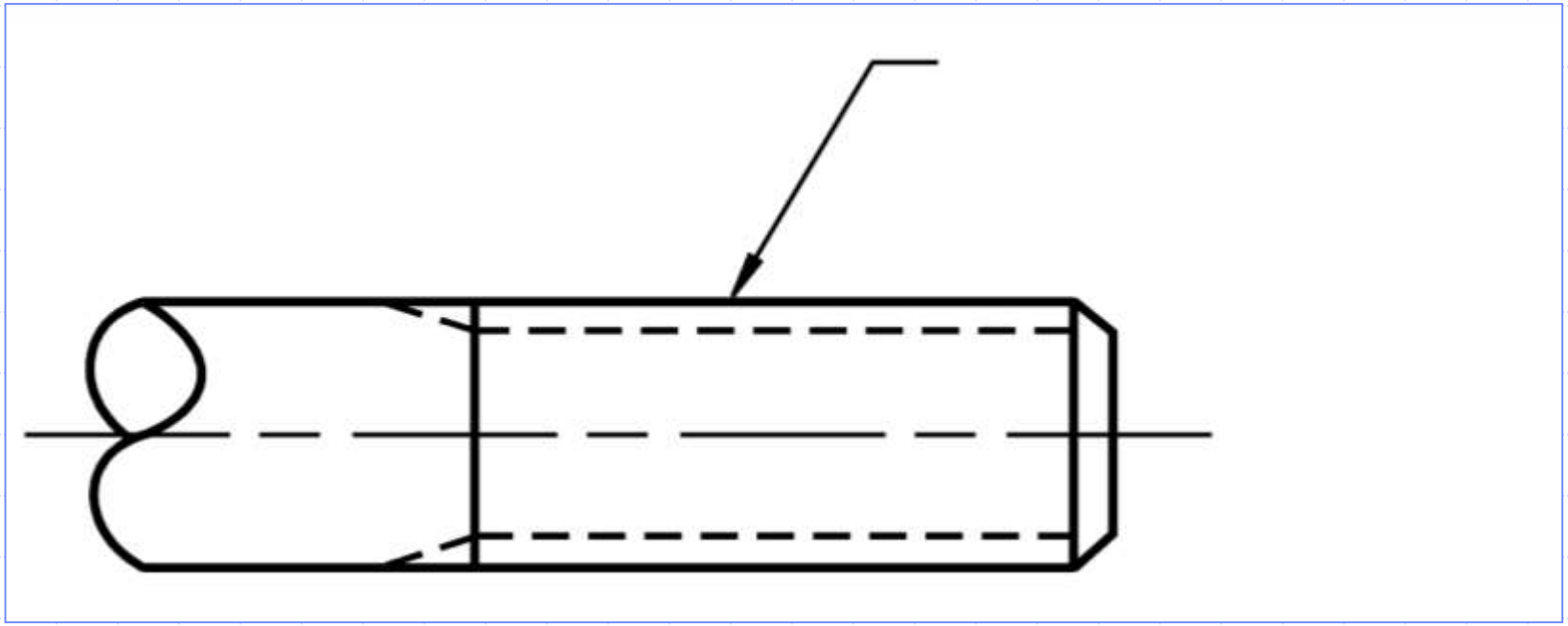
Exercise 10-3

Unified National thread note

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Exercise 10-3

- Write the thread note for a **#10 fine** thread. (See Appendix B)



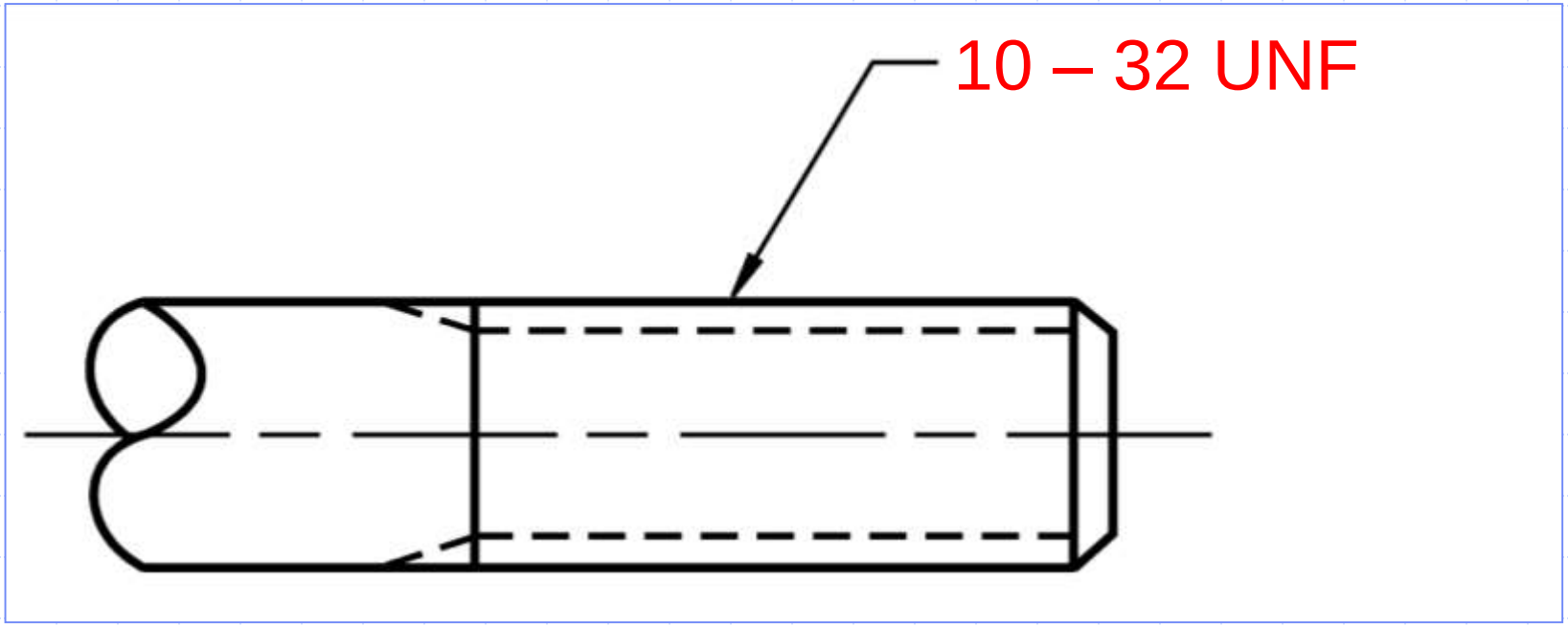
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(External Threads) Approximate Minor diameter = $D - 1.0825P$ P = Pitch

Nominal Size, in.	Basic Major Diameter (*D*)	Course UNC		Fine UNF		Extra Fine UNEF	
		Thds. Per in.	Tap Drill Dia.	Thds. Per in.	Tap Drill Dia.	Thds. Per in.	Tap Drill Dia.
#0	0.060	...	...	80	3/64	...	...
#1	0.0730	64	0.0595	72	0.0595	...	...
#2	0.0860	56	0.0700	64	0.0700	...	...
#3	0.0990	48	0.0785	56	0.0820	...	...
#4	0.1120	40	0.0890	48	0.0935	...	...
#5	0.1250	40	0.1015	44	0.1040	...	...
#6	0.1380	32	0.1065	40	0.1130	...	...
#8	0.1640	32	0.1360	36	0.1360	...	...
#10	0.1900	24	0.1495	32	0.1590	...	...
#12	0.2160	24	0.1770	28	0.1820	32	0.1850
1/4	0.2500	20	0.2010	28	0.2130	32	7/32
5/16	0.3125	18	0.257	24	0.272	32	9/32
3/8	0.3750	16	5/16	24	0.332	32	11/32
7/16	0.4375	14	0.368	20	25/64	28	13/32
1/2	0.5000	13	27/64	20	29/64	28	15/32
9/16	0.5625	12	31/64	18	33/64	24	33/64
5/8	0.6250	11	17/32	18	37/64	24	37/64
11/16	0.675	...	...	...	...	24	41/64
3/4	0.7500	10	21/32	16	11/16	20	45/64

Exercise 10-3

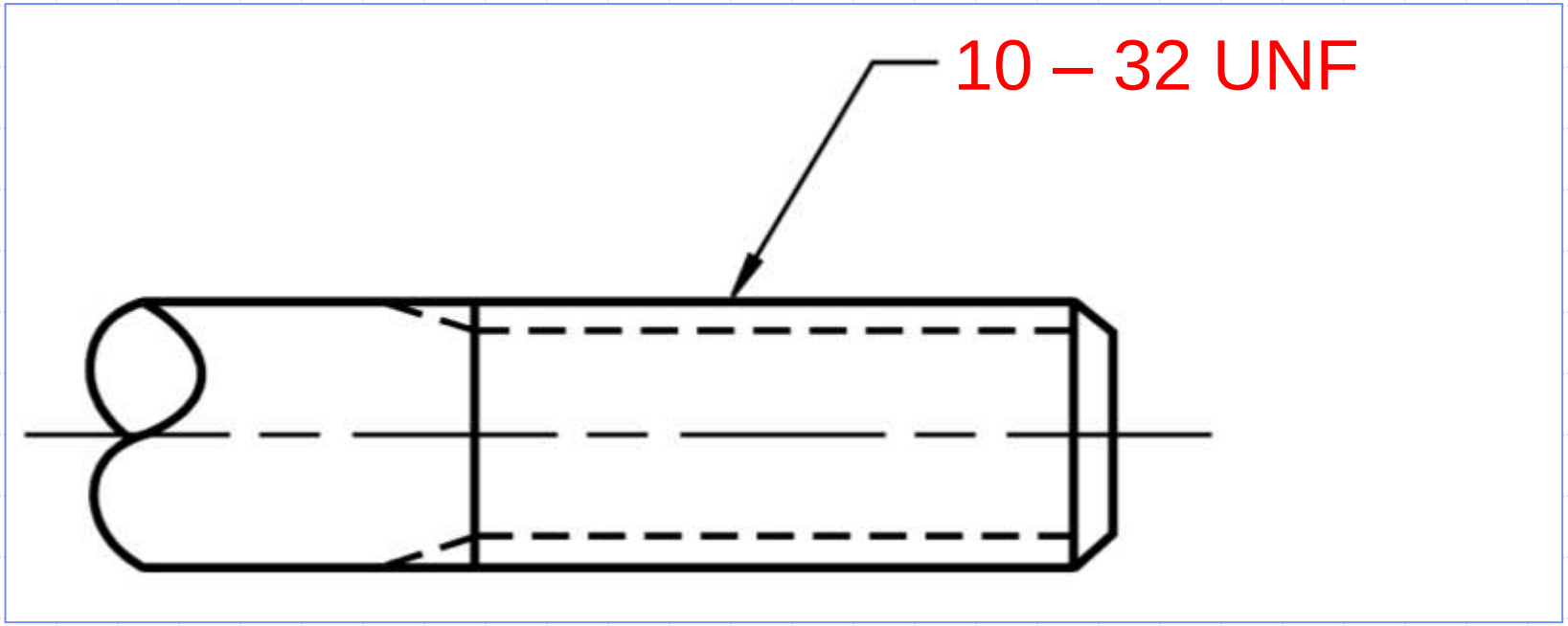
- Write the thread note for a **#10 fine** thread. (See Appendix B)



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Exercise 10-3

- Write the thread note for a **#10 fine** thread. (See Appendix B)
  - → Is the major diameter 10 inches? No



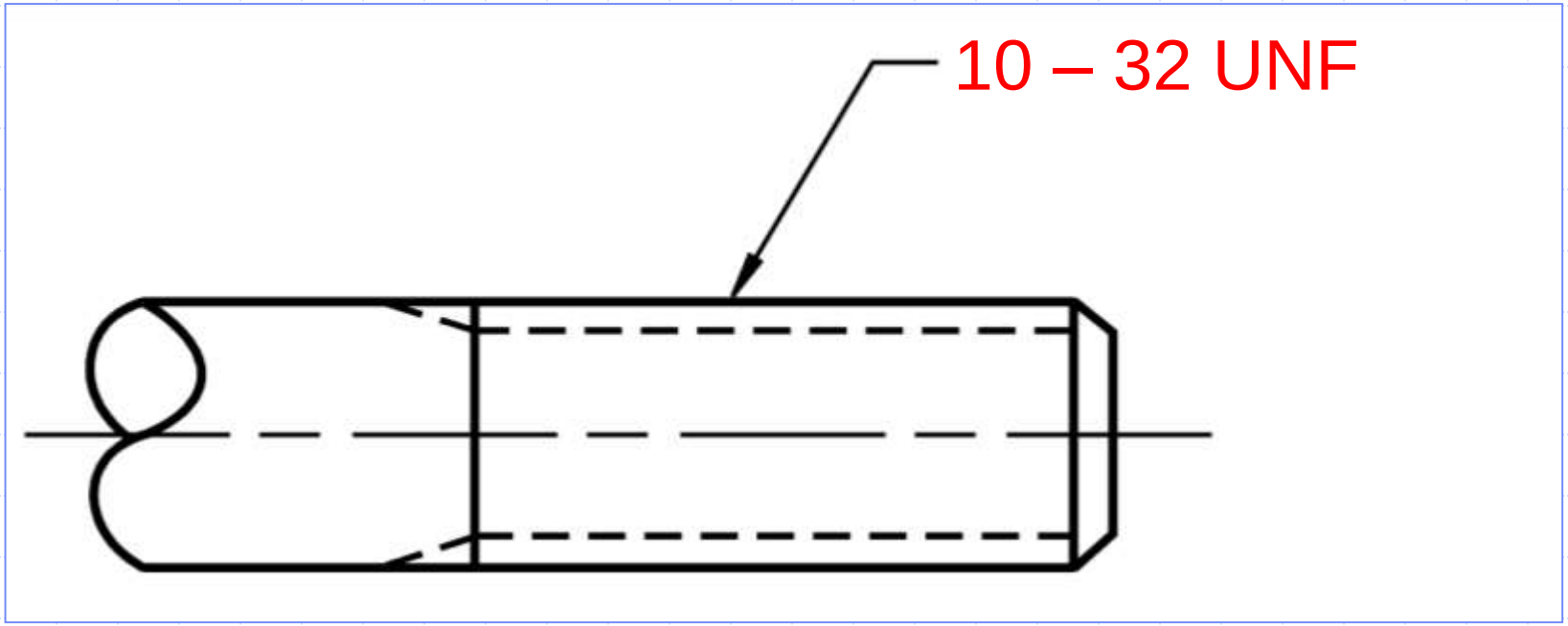
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(External Threads) Approximate Minor diameter = $D - 1.0825P$ P = Pitch

Nominal Size, in.	Basic Major Diameter (D)	Course UNC		Fine UNF		Extra Fine UNEF	
		Thds. Per in.	Tap Drill Dia.	Thds. Per in.	Tap Drill Dia.	Thds. Per in.	Tap Drill Dia.
#0	0.060	...	...	80	3/64	...	...
#1	0.0730	64	0.0595	72	0.0595	...	...
#2	0.0860	56	0.0700	64	0.0700	...	...
#3	0.0990	48	0.0785	56	0.0820	...	...
#4	0.1120	40	0.0890	48	0.0935	...	...
#5	0.1250	40	0.1015	44	0.1040	...	...
#6	0.1380	32	0.1065	40	0.1130	...	...
#8	0.1640	32	0.1360	36	0.1360	...	...
#10	0.1900	24	0.1495	32	0.1590	...	...
#12	0.2160	24	0.1770	28	0.1820	32	0.1850
1/4	0.2500	20	0.2010	28	0.2130	32	7/32
5/16	0.3125	18	0.257	24	0.272	32	9/32
3/8	0.3750	16	5/16	24	0.332	32	11/32
7/16	0.4375	14	0.368	20	25/64	28	13/32
1/2	0.5000	13	27/64	20	29/64	28	15/32
9/16	0.5625	12	31/64	18	33/64	24	33/64
5/8	0.6250	11	17/32	18	37/64	24	37/64
11/16	0.675	...	...	...	...	24	41/64
3/4	0.7500	10	21/32	16	11/16	20	45/64

Exercise 10-3

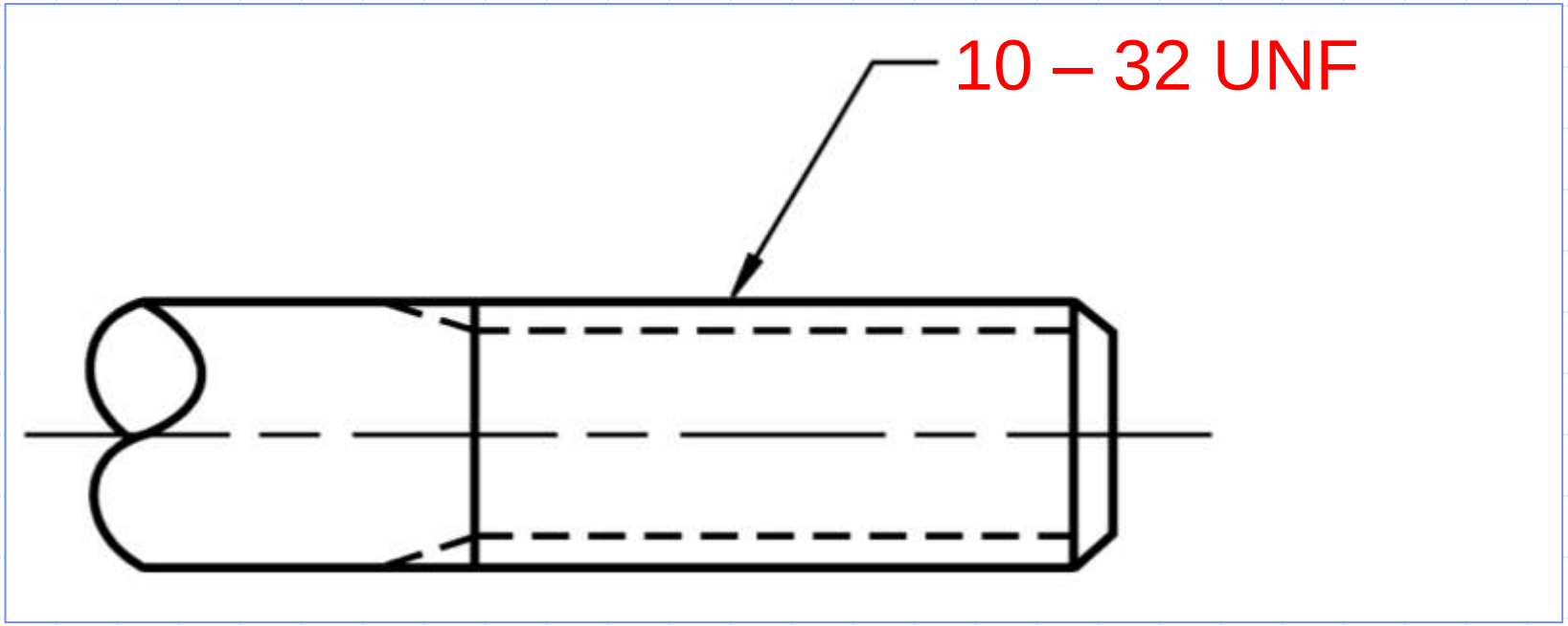
- Write the thread note for a **#10 fine** thread. (See Appendix B)
  - → Is the major diameter 10 inches? 0.190



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Exercise 10-3

- Write the thread note for a **#10 fine** thread. (See Appendix B)
  - → What is the minor diameter?



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(External Threads) Approximate Minor diameter = $D - 1.0825P$ P = Pitch

Nominal Size, in.	Basic Major Diameter (*D*)	Course UNC		Fine UNF		Extra Fine UNEF	
		Thds. Per in.	Tap Drill Dia.	Thds. Per in.	Tap Drill Dia.	Thds. Per in.	Tap Drill Dia.
#0	0.060	...	...	80	3/64	...	...
#1	0.0730	64	0.0595	72	0.0595	...	...
#2	0.0860	56	0.0700	64	0.0700	...	...
#3	0.0990	48	0.0785	56	0.0820	...	...
#4	0.1120	40	0.0890	48	0.0935	...	...
#5	0.1250	40	0.1015	44	0.1040	...	...
#6	0.1380	32	0.1065	40	0.1130	...	...
#8	0.1640	32	0.1360	36	0.1360	...	...
#10	0.1900	24	0.1495	32	0.1590	...	...
#12	0.2160	24	0.1770	28	0.1820	32	0.1850
1/4	0.2500	20	0.2010	28	0.2130	32	7/32
5/16	0.3125	18	0.257	24	0.272	32	9/32
3/8	0.3750	16	5/16	24	0.332	32	11/32
7/16	0.4375	14	0.368	20	25/64	28	13/32
1/2	0.5000	13	27/64	20	29/64	28	15/32
9/16	0.5625	12	31/64	18	33/64	24	33/64
5/8	0.6250	11	17/32	18	37/64	24	37/64
11/16	0.675	...	...	...	...	24	41/64
3/4	0.7500	10	21/32	16	11/16	20	45/64

Exercise 10-3

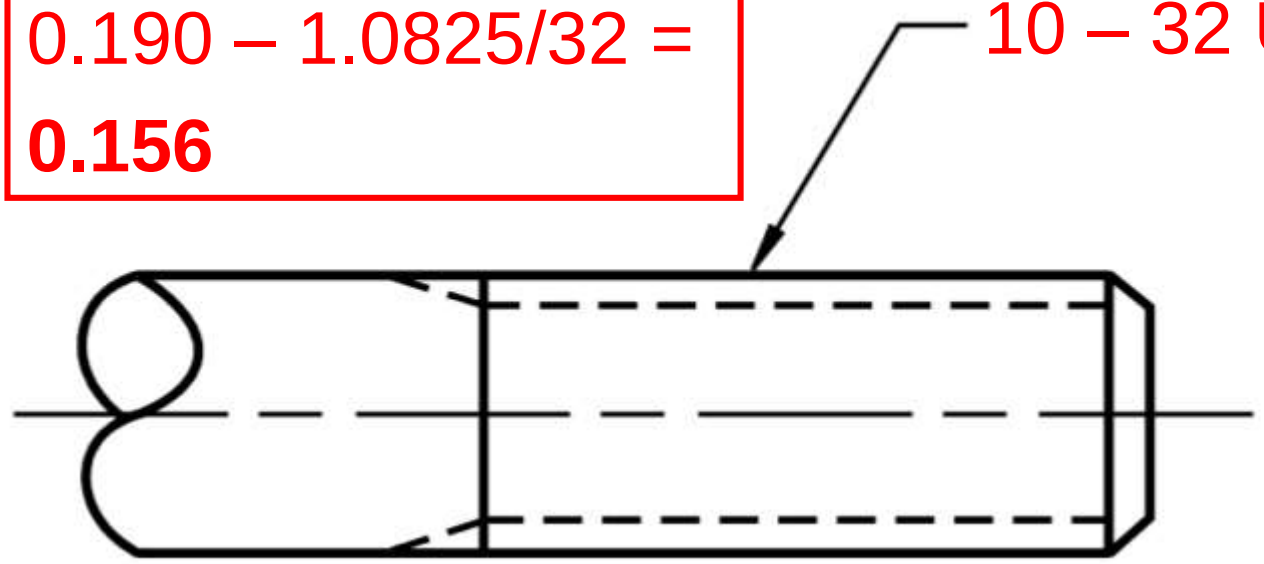
- Write the thread note for a **#10 fine** thread. (See Appendix B)

→ What is the minor diameter?

$D - 1.0825P =$

$0.190 - 1.0825/32 =$

0.156



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Threads and Fasteners

10.7) Metric Threads

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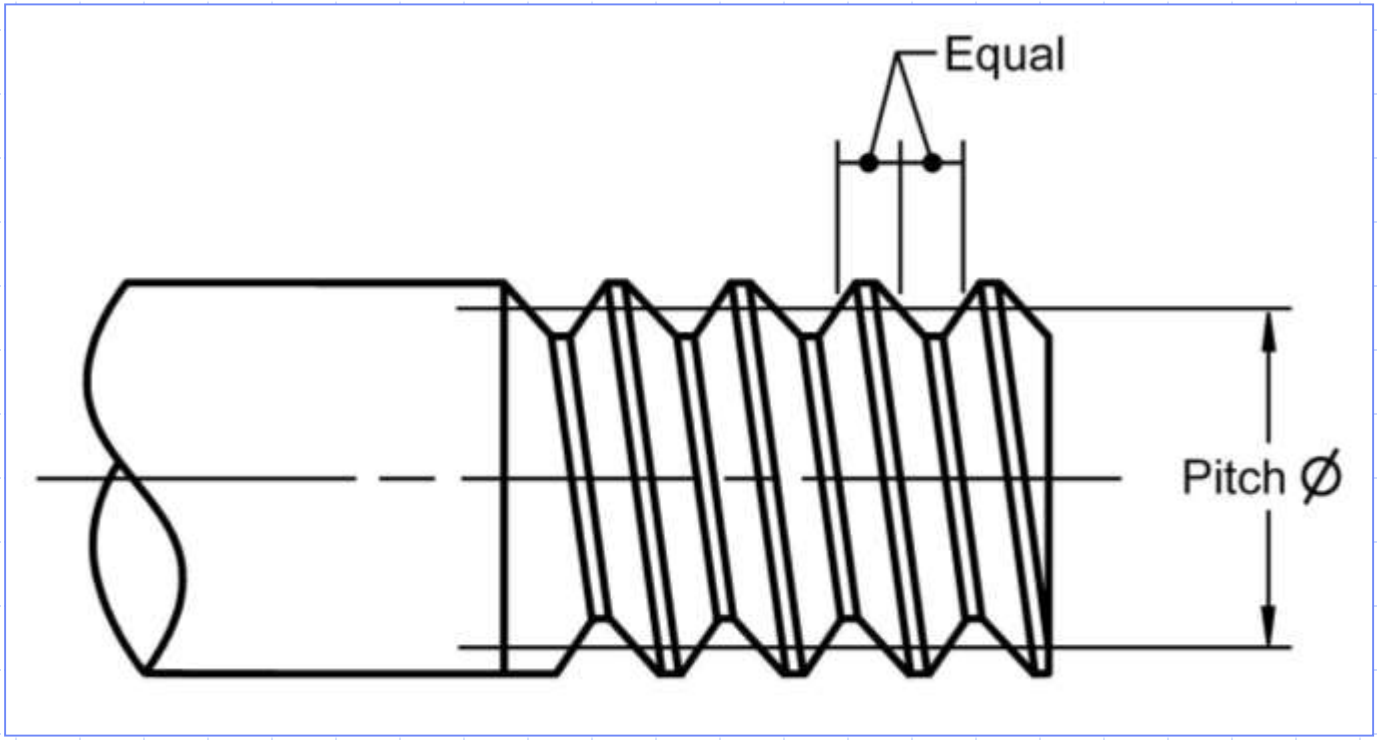
Metric Threads

- The metric thread note can contain a ***pitch diameter*** tolerance.
- What is the pitch diameter? Let's see.

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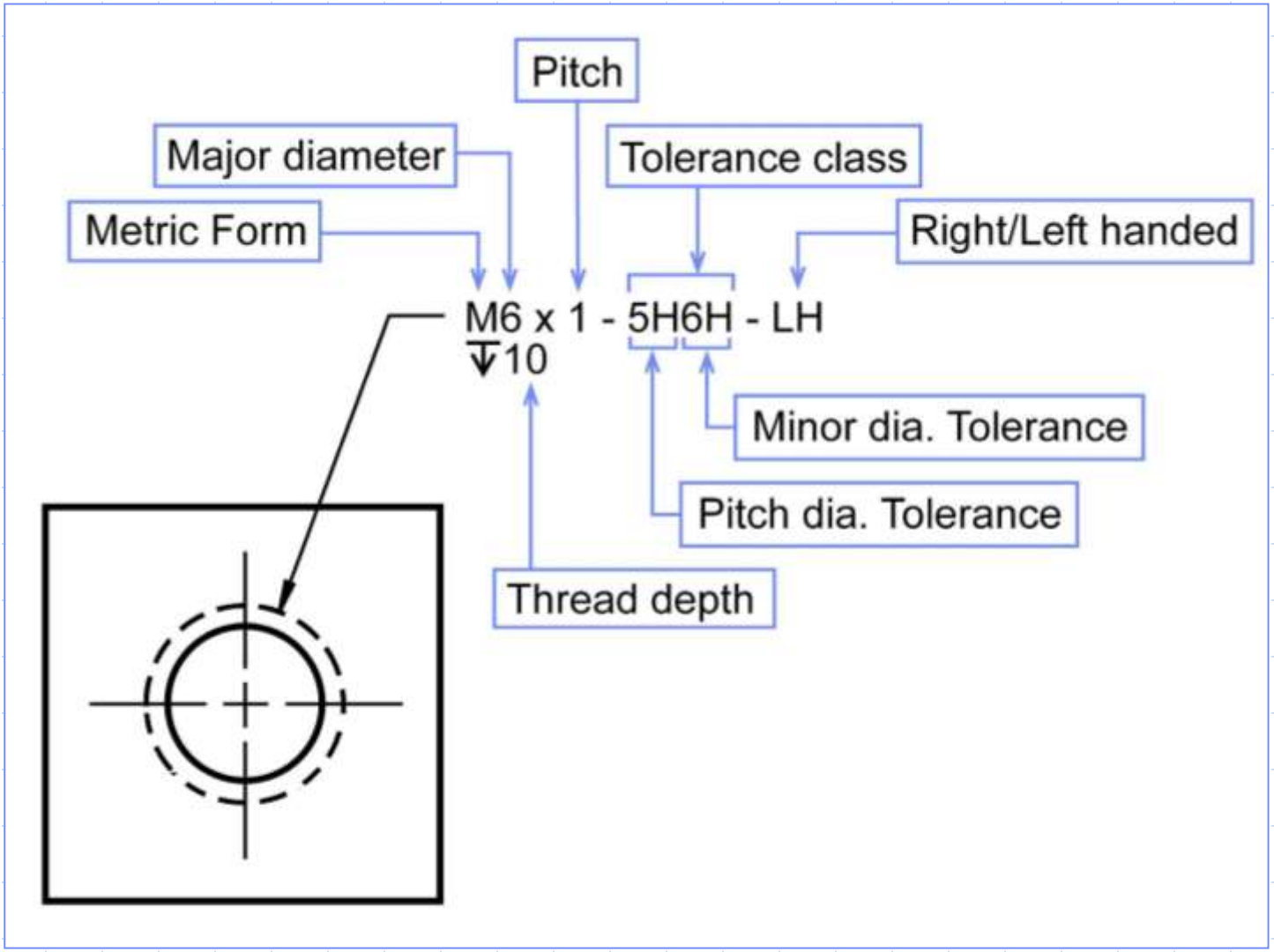
Pitch Diameter

- The *pitch diameter* cuts the threads at a point where the distance of the spaces equal the distance of the threads.



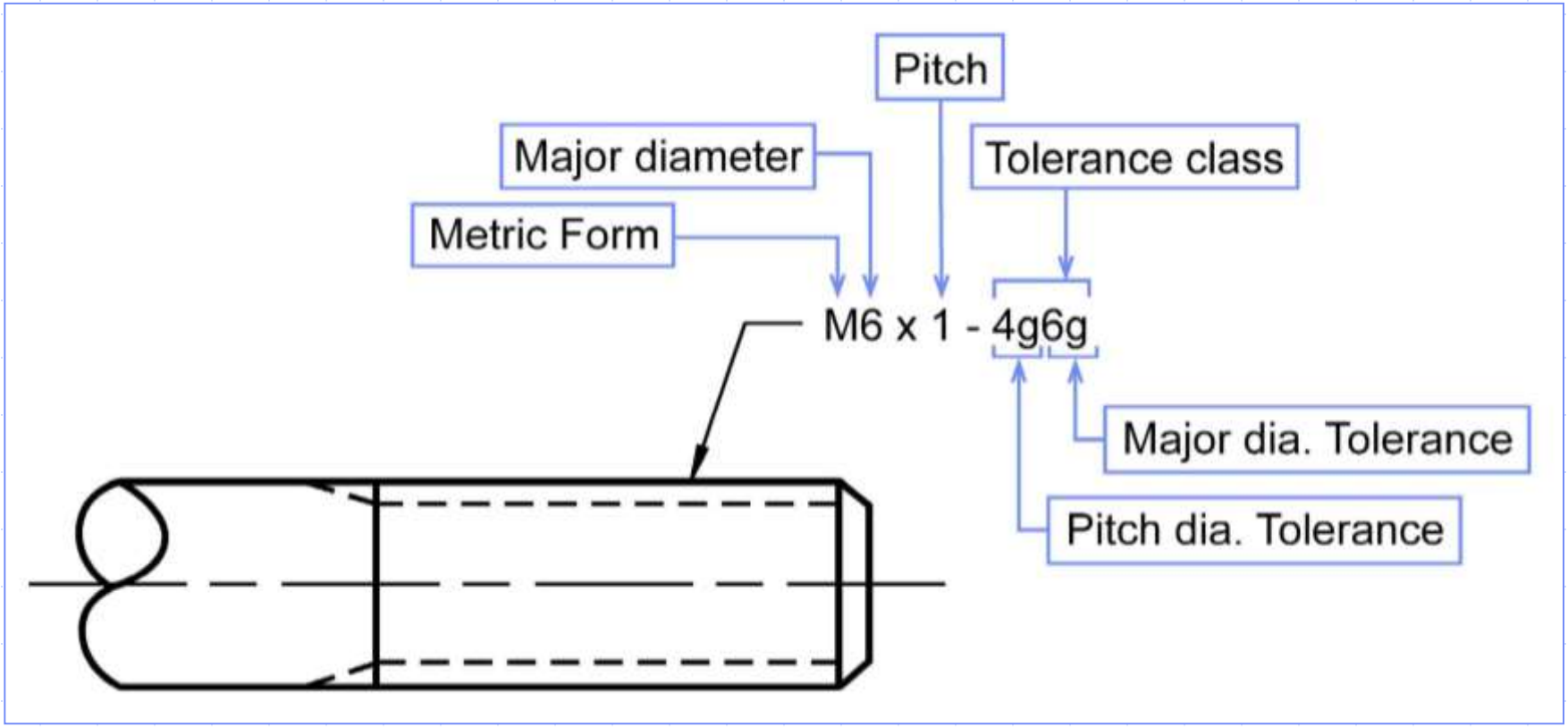
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Metric Thread Note Components



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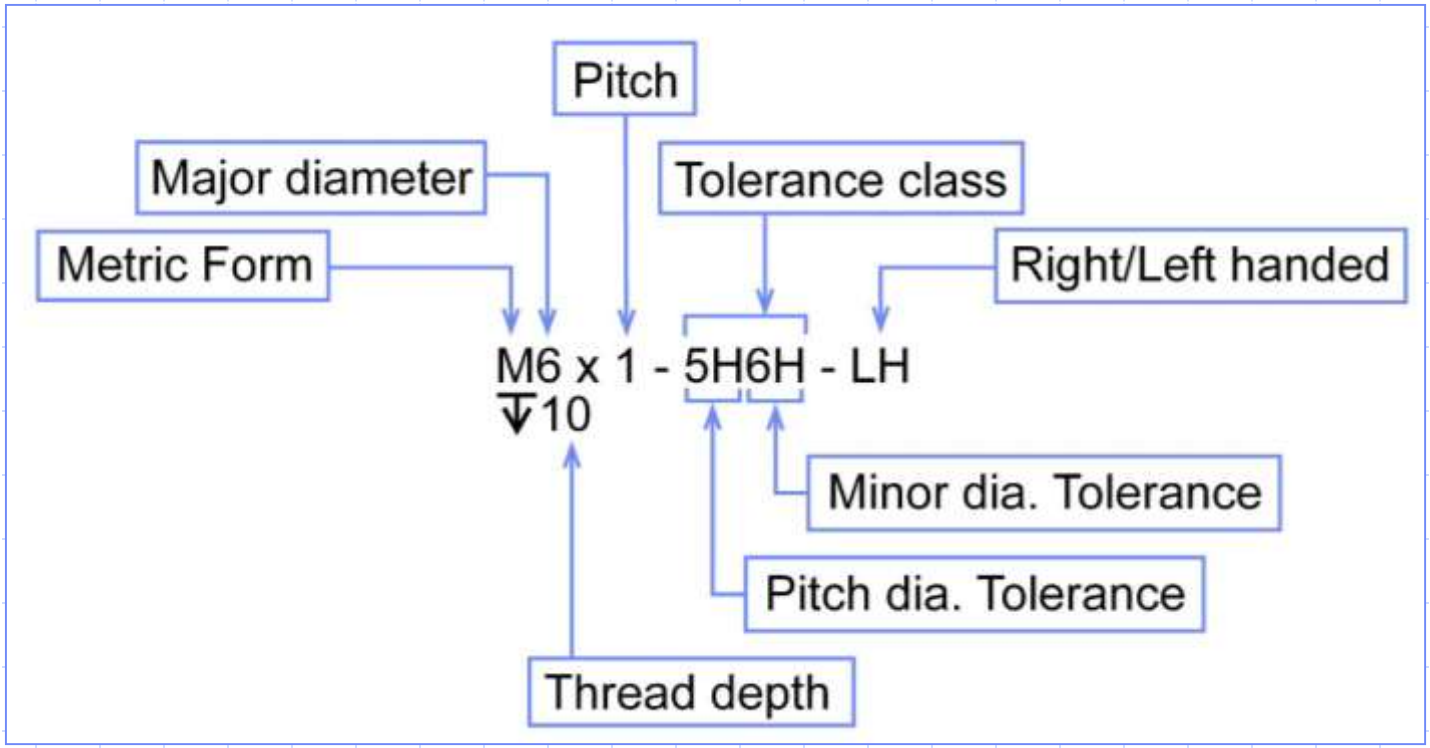
Metric Thread Note Components



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Metric Threads

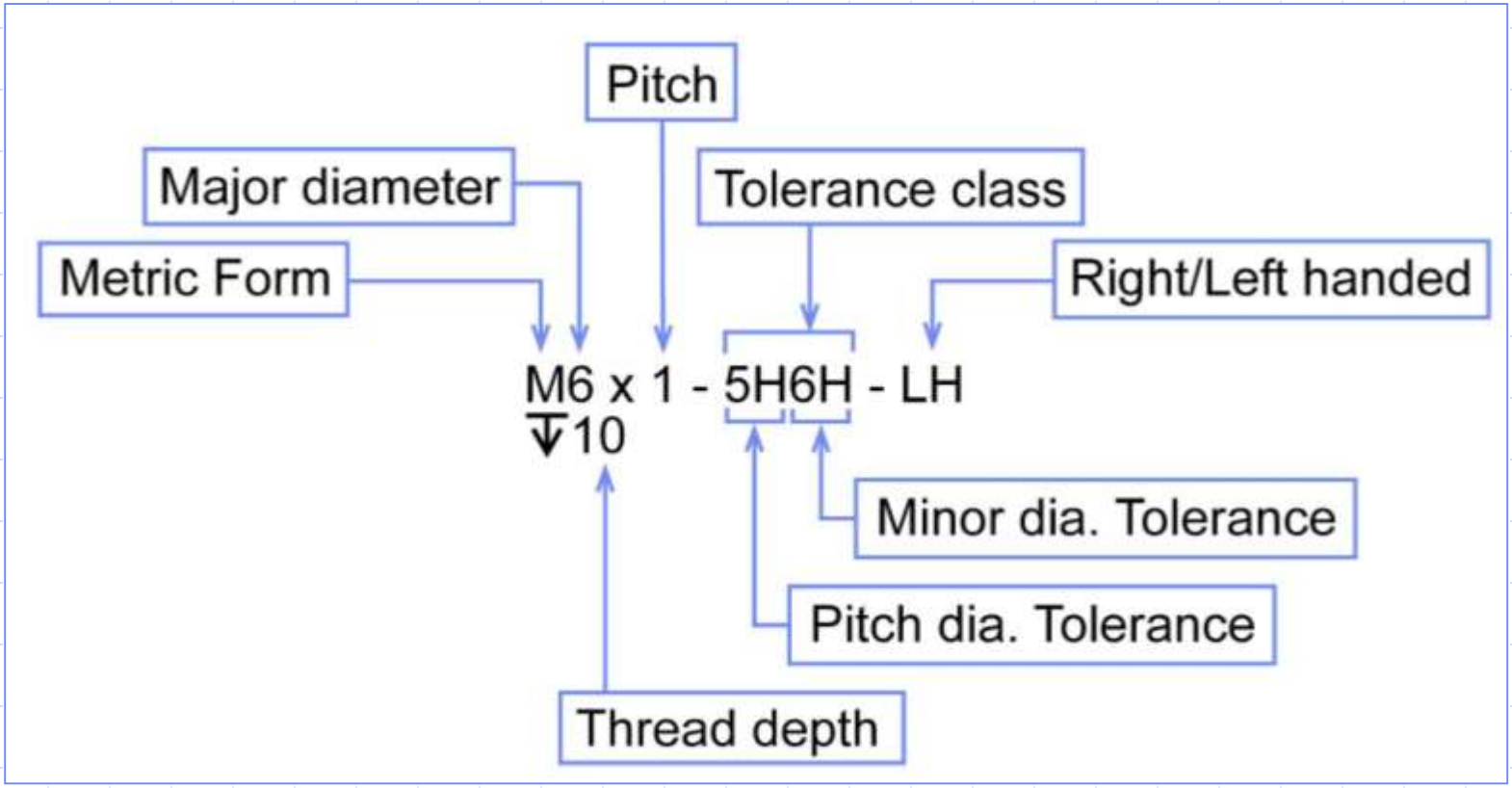
- **Metric Form:** Placing an **M** before the major diameter indicates the metric thread form.



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Metric Threads

- **Major Diameter:** The largest diameter
- **Pitch:** (P) Millimeters per thread.



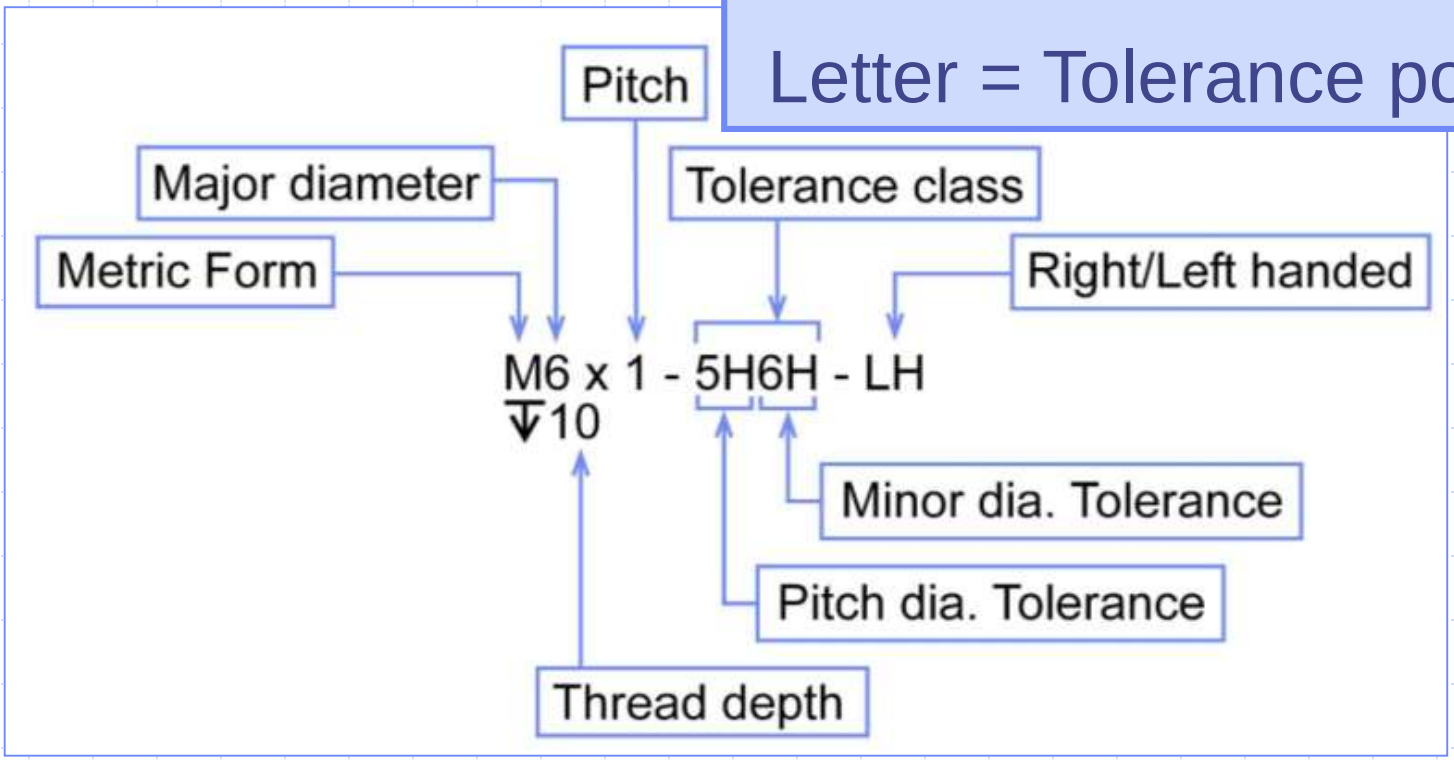
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Metric Threads

- **Tolerance Class:** It describes the looseness or tightness of fit between the internal and exte

Number = Tolerance grade

Letter = Tolerance position



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Metric Threads

- **Tolerance Class:**
  - → Tolerance Grade: Smaller numbers indicate a tighter fit.
  - → Tolerance Position: Specifies the amount of allowance.
    - Upper case letters = internal threads
    - Lower case letters = external threads.

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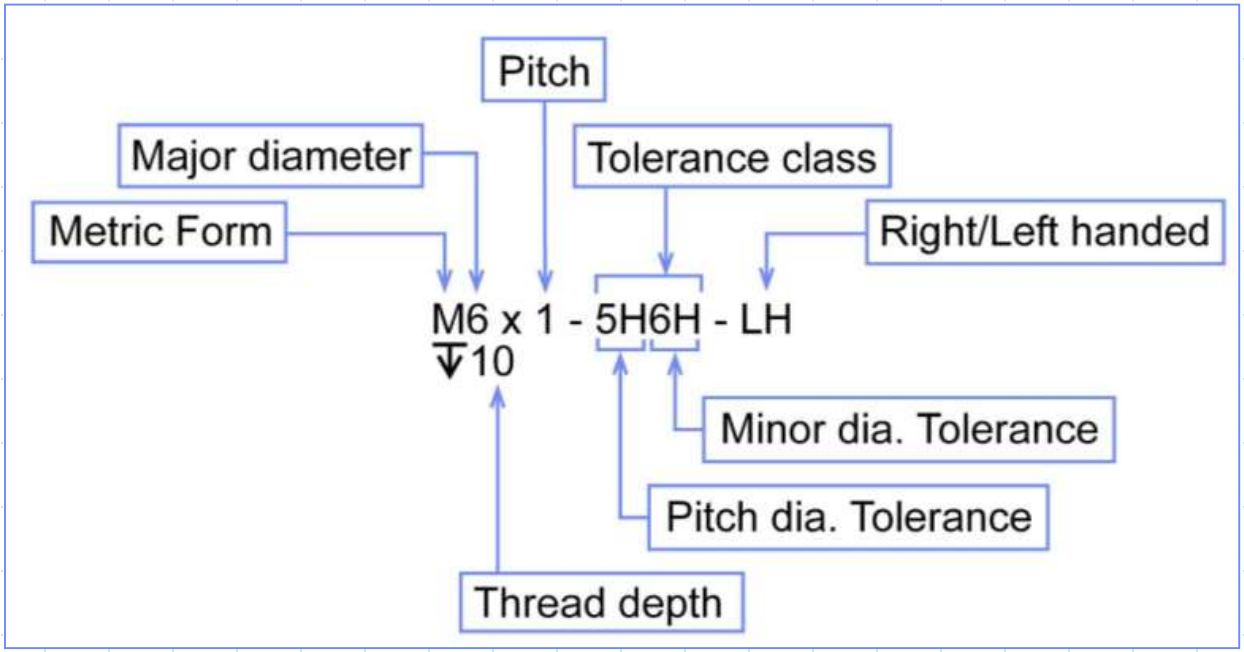
Metric Threads

- **<u>Tolerance Class:</u>** Two classes of metric thread fits are generally used.
  - → 6H/6g = General purpose
  - → 6H/5g6g = Closer fit.
  - → A tolerance class of 6H/6g is assumed if it is not specified.

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Metric Threads

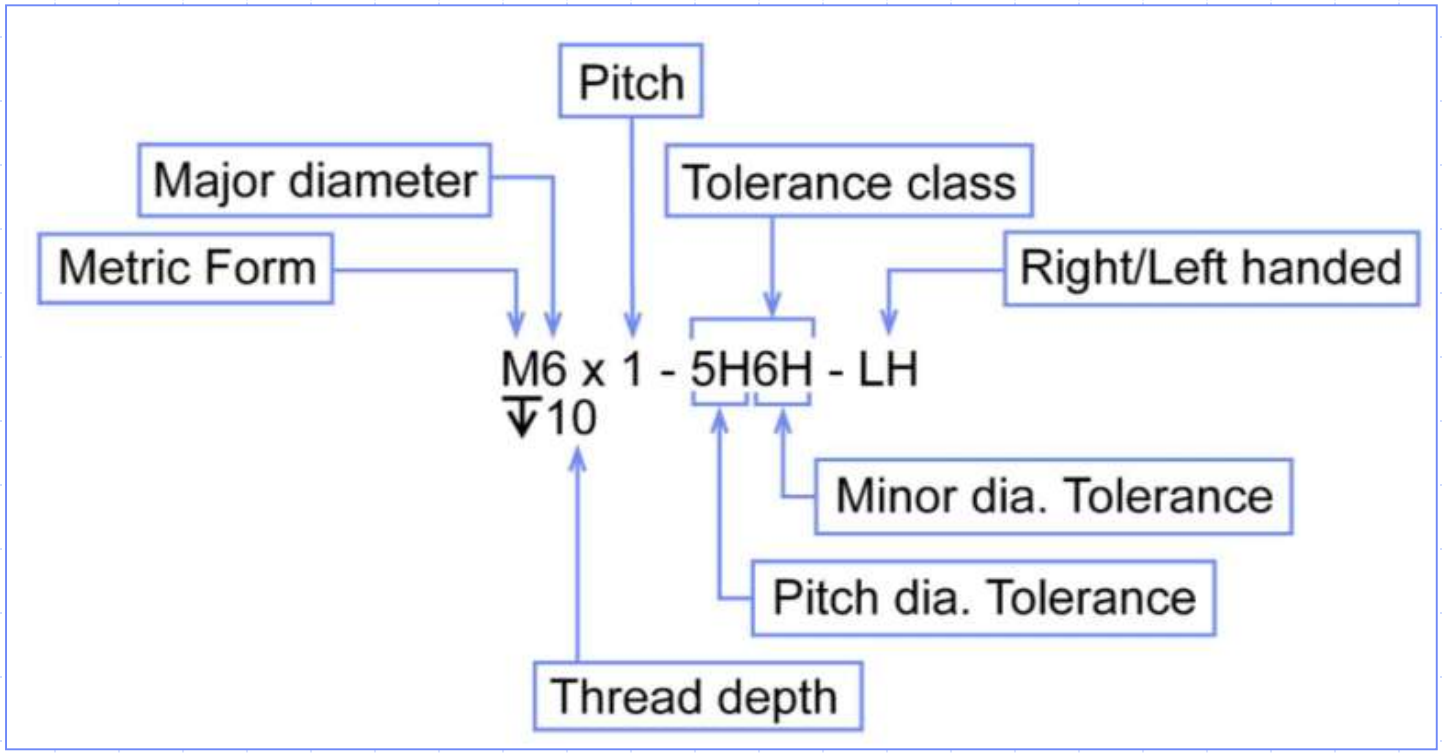
- **Right handed or Left handed thread:**
  - → **RH** = Right handed (right handed threads are assumed if not stated.)
  - → **LH** = Left handed



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Metric Threads

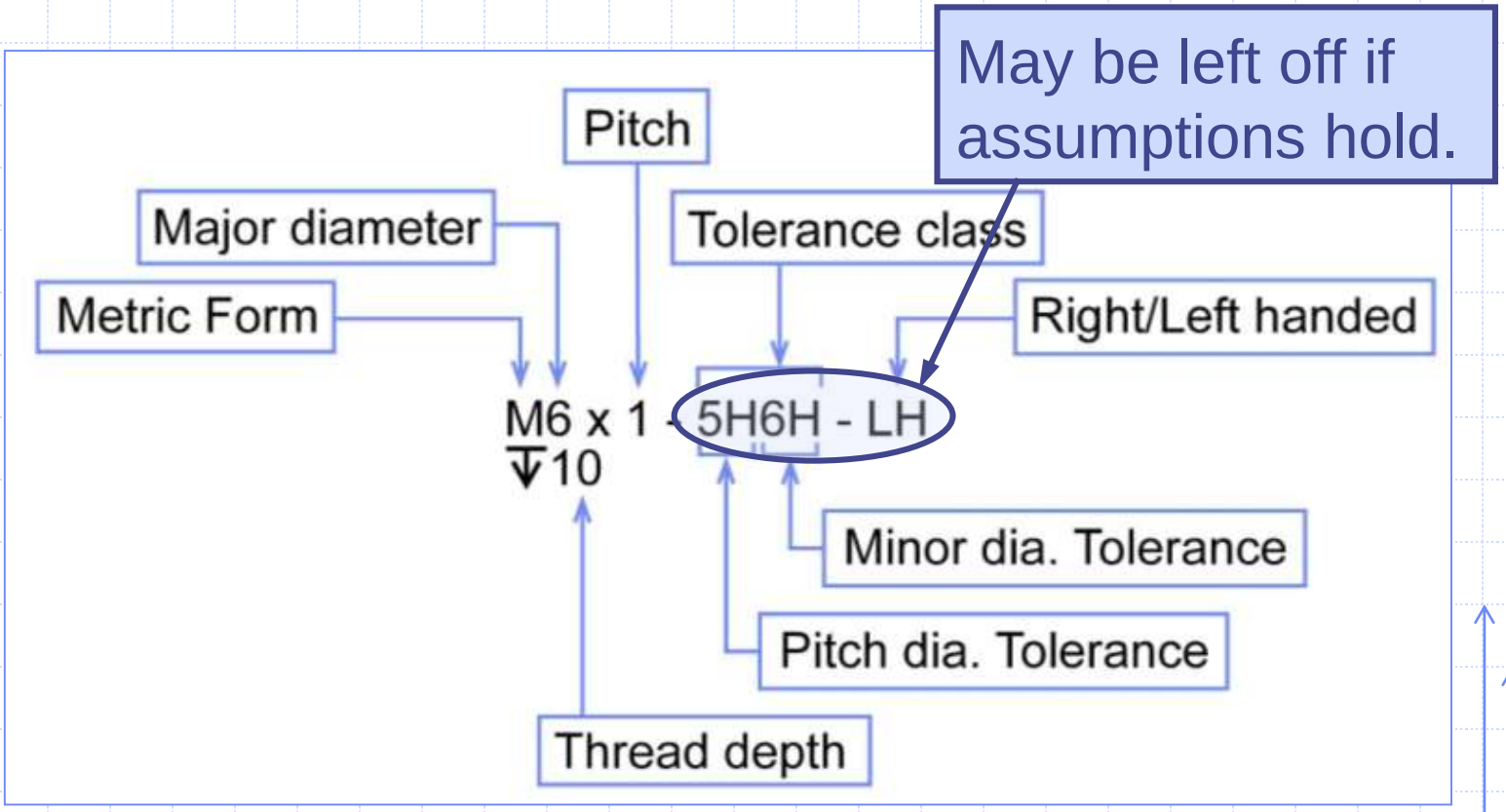
- **Depth of thread:** It indicates the thread depth for internal threads, not the tap drill depth.



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Metric Thread Note

- A tolerance class of **6H/6g** is assumed.
- Threads are assumed to be **RH**.



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Exercise 10-4

Metric thread note components

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Exercise 10-4

- Identify the different components of the following metric thread notes.
- **M10 x 1.5 – 4h6h – RH**

M	Metric Form
10	10 mm Major DIA
1.5	Pitch – mm/threads
4h	Pitch DIA tolerance
6h	Minor DIA tolerance
Int. or Ext.	External
RH	Right handed threads

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Metric Thread Tables

- **Standard screw thread tables are available in order to look up the;**
  - → Major diameter
  - → Pitch
  - → Tap drill size or Minor diameter

- Thread tables are located in Appendix B.

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Exercise 10-5

Metric thread tables

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Exercise 10-5

- For a **n16** internal metric thread, what are the;
  - → two available pitches,
  - → the tap drill diameter,
  - → and the corresponding minor diameter for the mating external threads.

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(External Threads) Approximate Minor diameter = $D - 1.2075P$ P = Pitch

Preferred sizes for commercial threads and fasteners are shown in boldface type.

Coarse (general purpose)		Fine	
Nominal Size (D) & Thread Pitch	**Tap Drill Diameter, mm**	**Nominal Size & Thread Pitch**	**Tap Drill Diameter, mm**
M1.6 x 0.35	1.25	---	---
M1.8 x 0.35	1.45	---	---
M2 x 0.4	1.6	---	---
M2.2 x 0.45	1.75	---	---
M2.5 x 0.45	2.05	---	---
M11 x 1.5	9.50	---	---
M12 x 1.75	10.30	**M12 x 1.25**	10.5
M14 x 2	12.00	**M14 x 1.5**	12.5
M16 x 2	14.00	**M16 x 1.5**	14.5
M18 x 2.5	15.50	**M18 x 1.5**	16.5

Find this page.

Exercise 10-5

- For a **n16** internal metric thread.

Pitch	**Tap drill DIA**	**Minor DIA** (External)
2	14	13.6
1.5	14.5	14.2

(External Threads) Approximate Minor diameter = $D - 1.2075P$ P = Pitch

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Exercise 10-5

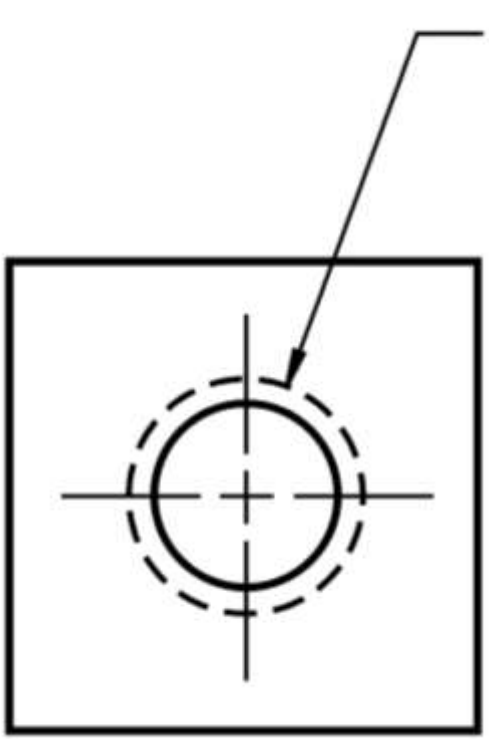
- For a **n16** internal metric thread.

- Which has the finer thread?
  - → Pitch = 2
  - → Pitch = 1.5

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Exercise 10-5

- Write the thread note for a **16** mm diameter **coarse** thread.



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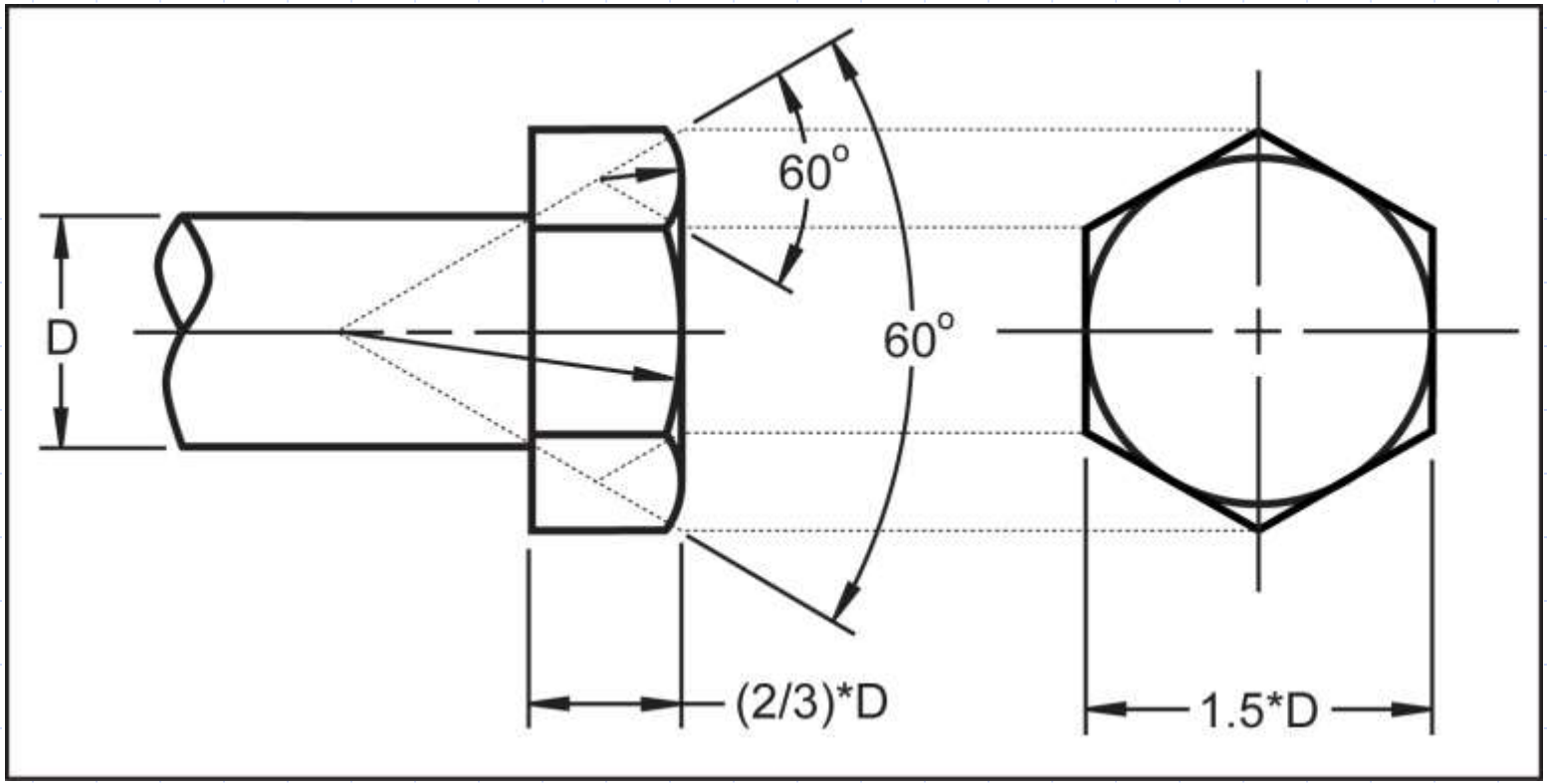
Threads and Fasteners

10.8) Drawing Bolts

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Drawing Bolts

- **D** represents the major diameter.
- Nuts are drawn in a similar fashion.



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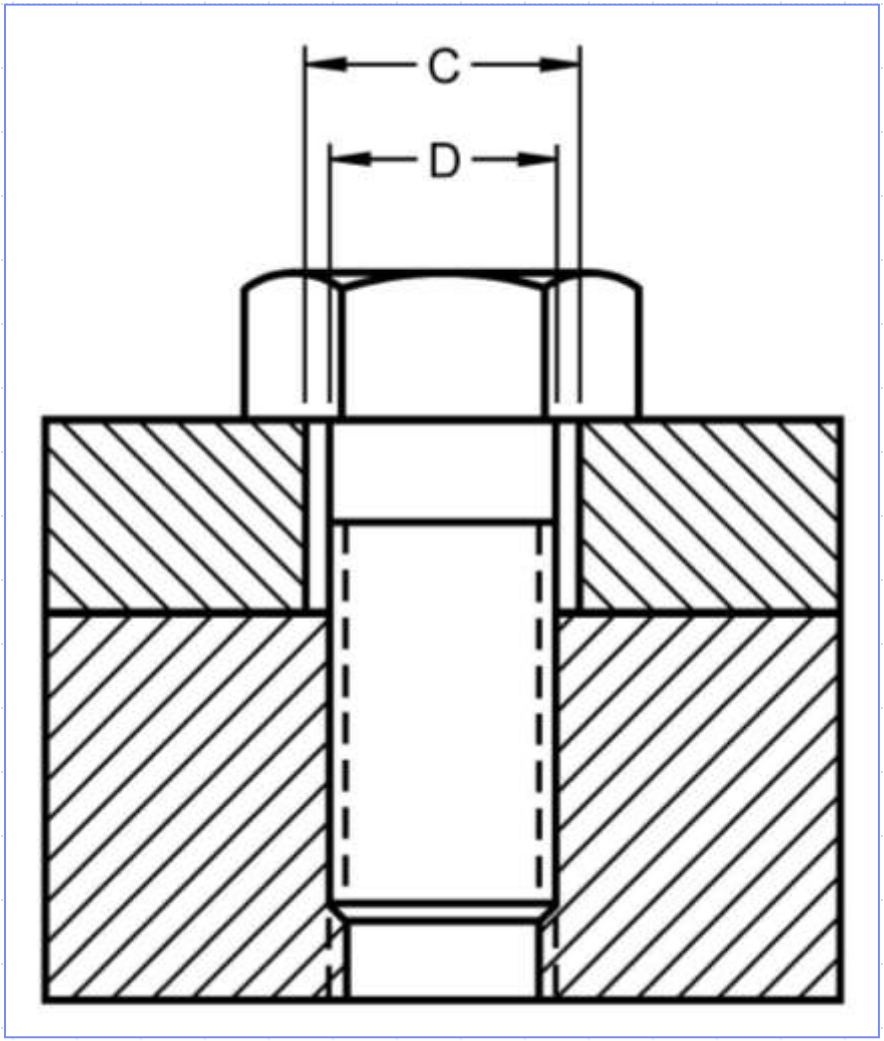
Threads and Fasteners

10.9) Bolt and Screw Clearances

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Bolt and Screw Clearances

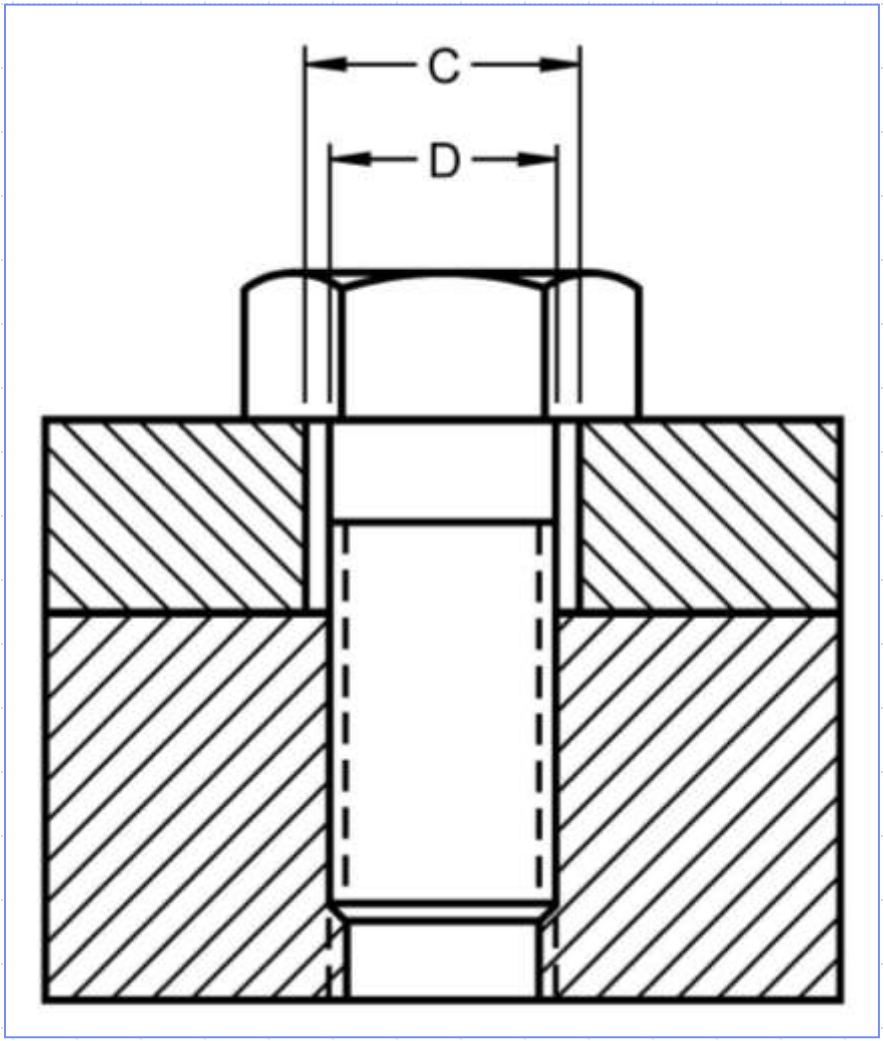
- Bolts and screws attach one material with a clearance hole to another material with a threaded hole.



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Bolt and Screw Clearances

- The size of the clearance hole depends on;
  - → the major diameter of the fastener
  - → and the type of fit
    - normal
    - close
    - loose



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Table 10-2 (Normal fit clearances)

- Other fits may be found in Appendix B.

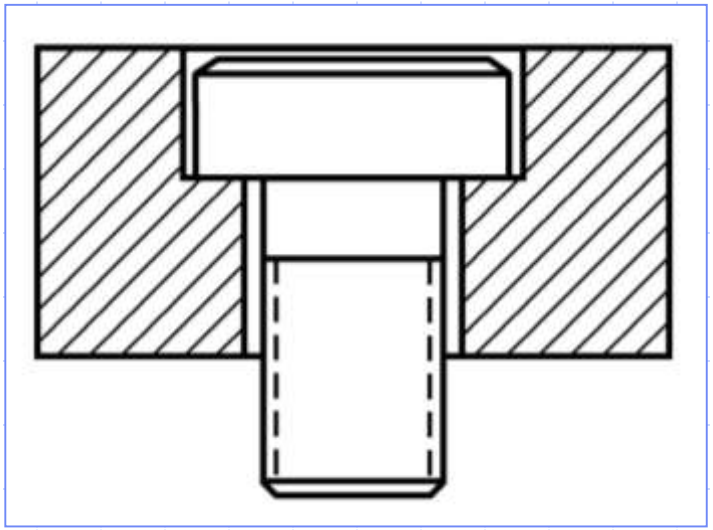
Inch clearances	
Nominal screw size (D)	Clearance hole (C)
#0 - #4	D + 1/64
#5 – 7/16	D + 1/32
1/2 – 7/8	D + 1/16
1	D + 3/32
1 1/8, 1 1/4	D + 3/32
1 3/8, 1 1/2	D + 1/8

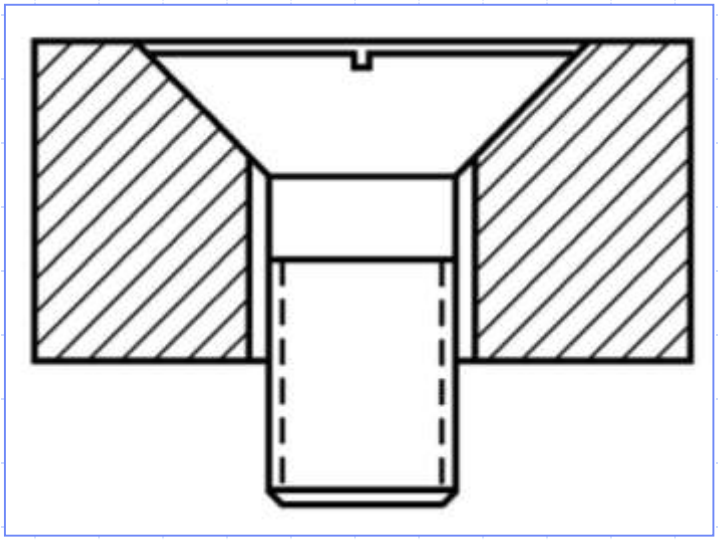
Metric clearances	
Nominal screw size (D)	Clearance hole (C)
M1.6	D + 0.2
M2, M2.5	D + 0.4
M4, M5	D + 0.5
M6	D + 0.6
M8, M10	D + 1
M12 – M16	D + 1.5
M20, M24	D + 2
M30 – M42	D + 3
M48	D + 4
M56 – M90	D + 6
M100	D + 7

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Bolt and Screw Clearances

- Sometimes bolt or screw heads need to be flush with the surface. This can be achieved by using either a **counterbore** or **countersink** depending on the fasteners head shape.



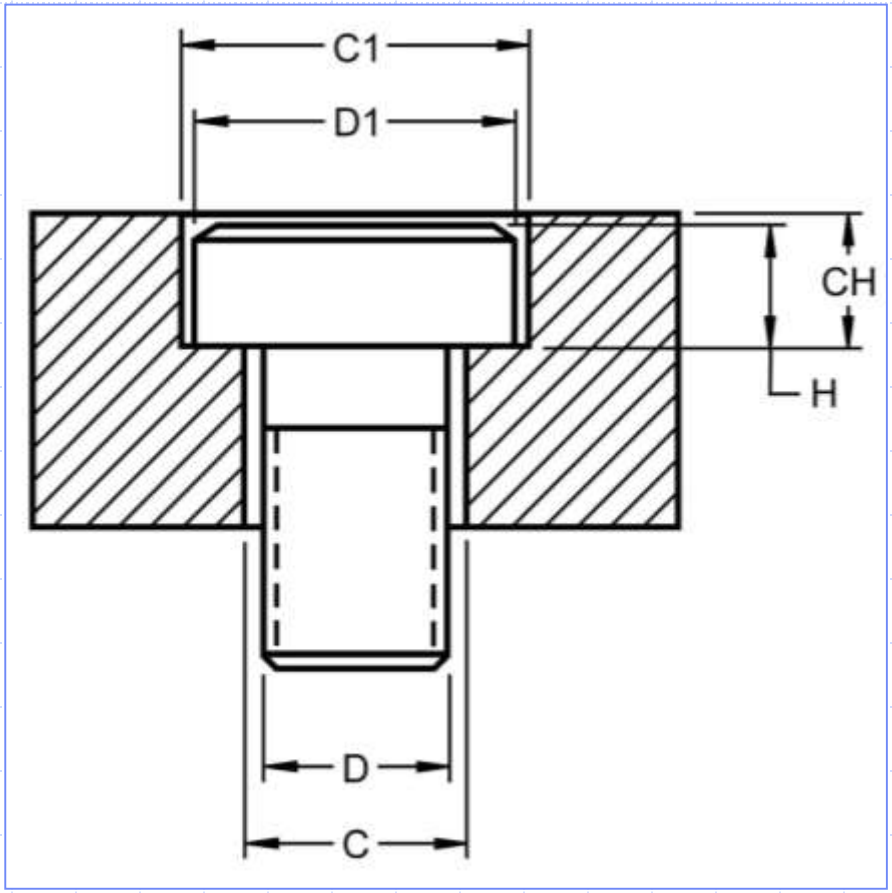


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Bolt and Screw Clearances

- **Counterbores:**
  Counterbores are holes designed to recess bolt or screw heads below the surface of a part.

Typically,
CH = H + 1/16 (1.5 mm)
and
C1 = D1 + 1/8 (3 mm)



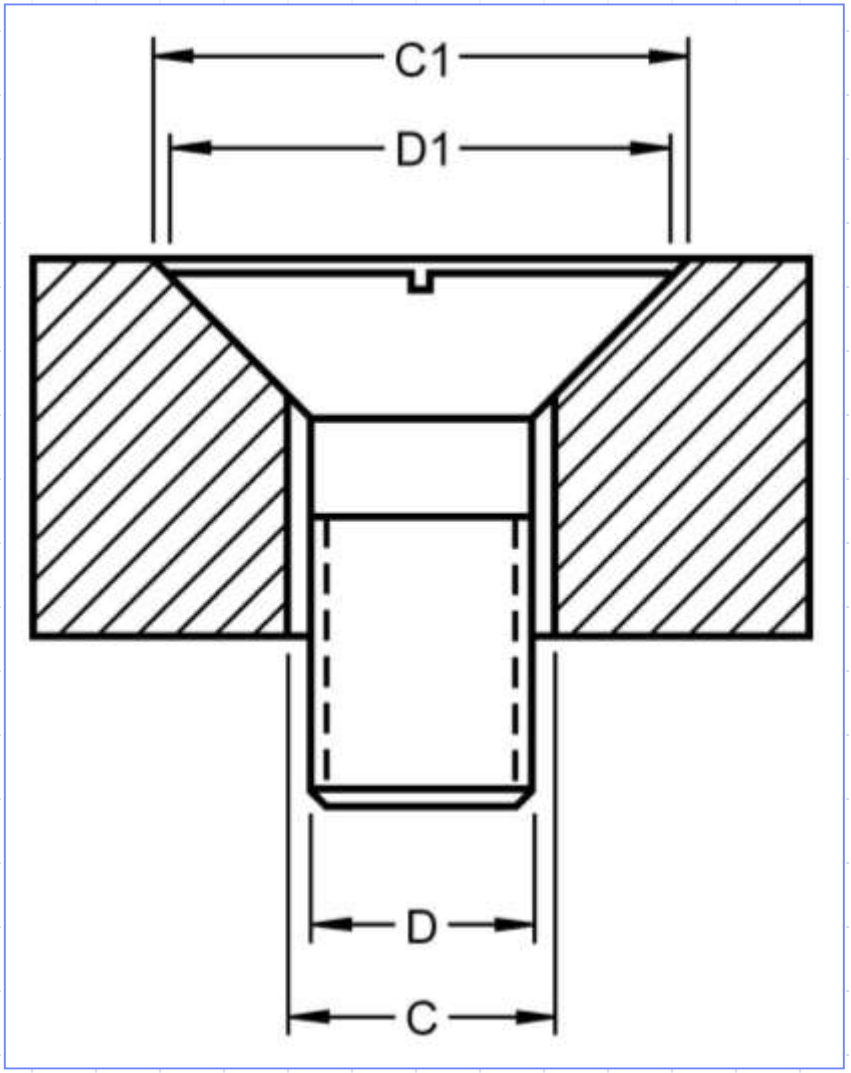
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Bolt and Screw Clearances

- **Countersink:**
  Countersinks are angled holes that are designed to recess screws with angled heads.

Typically,
C1 = D1 + 1/8 (3 mm)

Appendix B gives other counterbore, countersink and shaft clearance holes.



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Exercise 10-6

Fastener tables and clearance holes

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Exercise 10-6

- What is the normal fit clearance hole diameter for the following nominal bolt sizes.

Nominal size	Clearance hole
1/4	9/32
3/4	13/16

Inch clearances	
Nominal screw size (D)	Clearance hole (C)
#0 - #4	D + 1/64
#5 – 7/16	D + 1/32
1/2 – 7/8	D + 1/16
1	D + 3/32
1 1/8, 1 1/4	D + 3/32
1 3/8, 1 1/2	D + 1/8

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Exercise 10-6

- A **5/16 - 18 UNC – Socket Head Cap Screw** needs to go through a piece of metal in order to screw into a plate below.
- The head of the screw should be flush with the surface.

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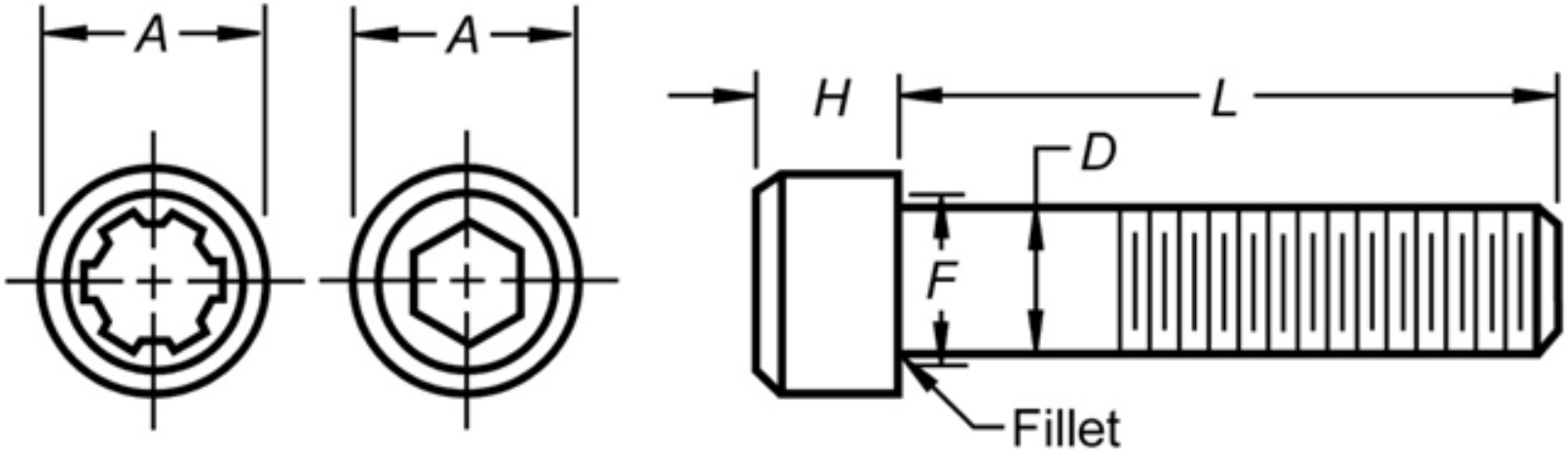
Exercise 10-6

- **5/16 - 18 UNC – Socket Head Cap Screw**
- Fill in the following table. Refer to Appendix B.

Head diameter	
Height of head	
Normal clearance hole dia.	
C’Bore dia.	
C’Bore depth	

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B.3.3) Dimensions of Hexagon and Spline Socket Head Cap Screws



Screw Size (*D*)	Head Diameter	Head Height
#0 - #10	See Table	Max. *H* = *D*
1/4 - 4	Max. *A* = 1.50 *D*	

D = 5/16

Screw Size (*D*)	#0	#1	#2	#3	#4
Max. Head Diameter (*A*)	0.096	0.118	0.140	0.161	0.183

Screw Size (*D*)	#5	#6	#8	#10
Max. Head Diameter (*A*)	0.205	0.226	0.270	0.312

Exercise 10-6

- **5/16 - 18 UNC – Socket Head Cap Screw**
- Fill in the following table. Refer to Appendix B.

Max. Head diameter	**A = 1.5(5/16)=0.469**
Max. Height of head	**H = D = 5/16**
Normal clearance hole dia.	
C'Bore dia.	
C'Bore depth	

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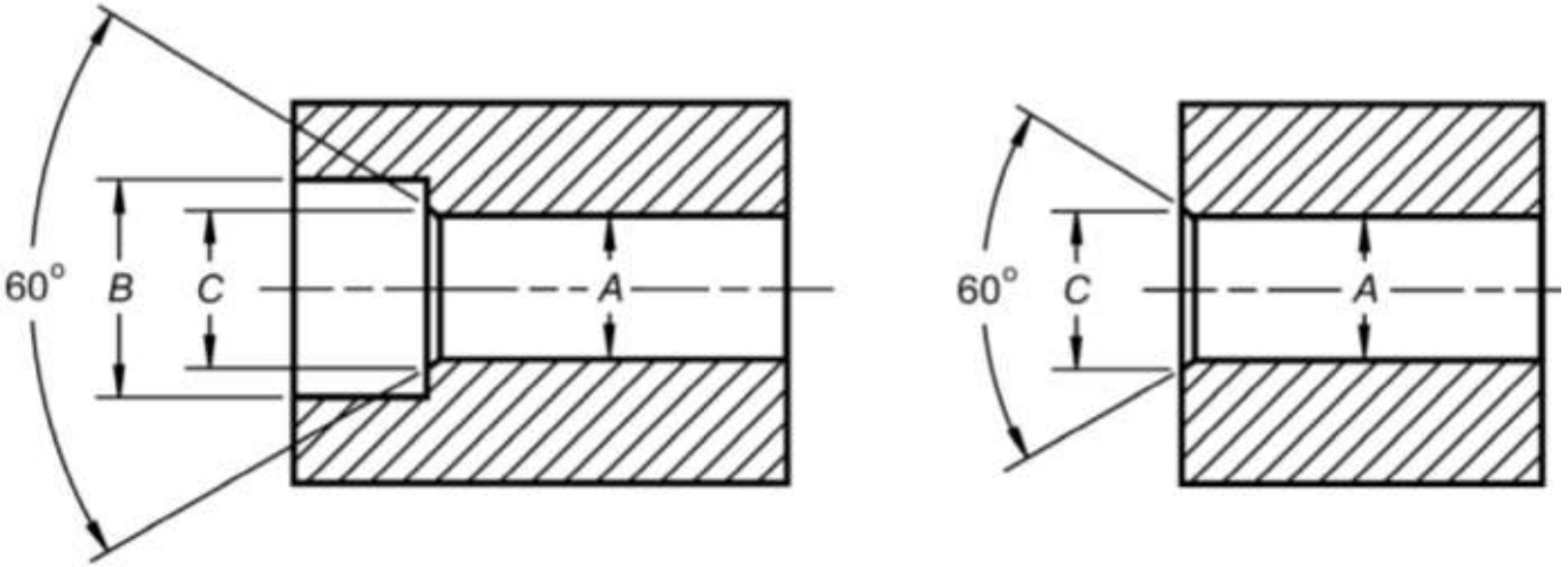
Exercise 10-6

- **5/16 - 18 UNC – Socket Head Cap Screw**
- Fill in the following table. Refer to Appendix B.

Max. Head diameter	A = 1.5(5/16)=.469
Max. Height of head	H = D = 5/16
Normal clearance hole dia.	
C'Bore dia.	
C'Bore depth	

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Drill and Counterbore Sizes for Socket Head Cap Screws



Nominal Size of Screw (*D*)	Nominal Drill Size (*A*)		Counterbore Diameter (*B*)	Countersink (*C*)
	Close Fit	Normal Fit		
#0 (0.0600)	(#51) 0.067	(#49) 0.073	1/8	0.074
#1 (0.0730)	(#46) 0.081	(#43) 0.089	5/32	0.087
#10 (0.1900)	(#5) 0.206	(#2) 0.221	3/8	0.218
1/4	17/64	9/32	7/16	0.278
5/16	21/64	11/32	17/32	0.346
3/8	25/64	13/32	5/8	0.415

Exercise 10-6

- **5/16 - 18 UNC – Socket Head Cap Screw**
- Fill in the following table. Refer to Appendix B.

Max. Head diameter	A = 1.5(5/16)=.469
Max. Height of head	H = D = 5/16
Normal clearance hole dia.	**C = D + 1/32 = 11/32**
C'Bore dia.	**B = 17/32**
C'Bore depth	

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Exercise 10-6

- **5/16 - 18 UNC – Socket Head Cap Screw**
- Fill in the following table. Refer to Appendix B.

Max. Head diameter	A = 1.5(5/16)=.469
Max. Height of head	H = D = 5/16
Normal clearance hole dia.	C = D + 1/32 = 11/32
C'Bore dia.	B = 17/32
C'Bore depth	

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Exercise 10-6

- **5/16 - 18 UNC – Socket Head Cap Screw**
- Fill in the following table. Refer to Appendix B.

Max. Head diameter	A = 1.5(5/16)=.469
Max. Height of head	H = D = 5/16
Normal clearance hole dia.	C = D + 1/32 = 11/32
C'Bore dia.	B = 17/32
C'Bore depth	**>H (H+1/16 = 3/8)**

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Exercise 10-6

- An **M8x1.25 Flat Countersunk Head Metric Cap Screw** needs to go through a piece of metal in order to screw into a plate below.
- The clearance hole needs to be close and the head needs to be flush with the surface.
- What should the countersink diameter and clearance hole diameter be?

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Exercise 10-6

- **M8x1.25 Flat Countersunk Head Metric Cap Screw**

Major dia.	
Head dia.	
C’Sink dia.	
Close clearance hole dia.	

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Exercise 10-6

- **M8x1.25 Flat Countersunk Head Metric Cap Screw**

Major dia.	**8**
Head dia.	
C’Sink dia.	
Close clearance hole dia.	

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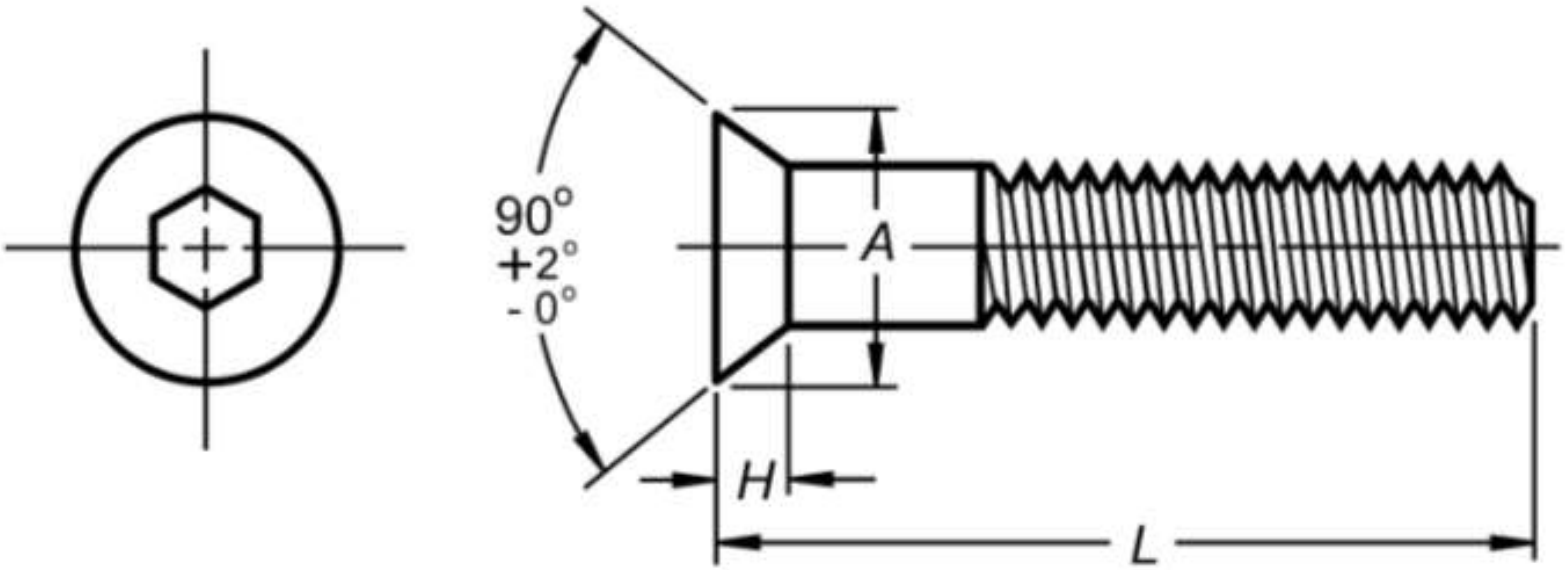
Exercise 10-6

- **M8x1.25 Flat Countersunk Head Metric Cap Screw**

Major dia.	8
Head dia.	
C'Sink dia.	
Close clearance hole dia.	

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Dimensions of Metric Countersunk Socket Head Cap Screws



Basic Screw Diameter and Thread Pitch	Head Diameter (*A*) Theor. Sharp	Head Height (*H*)
M3 x 0.5	6.72	1.86
M4 x 0.7	8.96	2.48
M5 x 0.8	11.20	3.10
M6 x 1	13.44	3.72
M8 x 1.25	17.92	4.96
M10 x 1.5	22.40	6.20

Exercise 10-6

- **M8x1.25 Flat Countersunk Head Metric Cap Screw**

Major dia.	8
Head dia.	**A = 17.92**
C'Sink dia.	
Close clearance hole dia.	

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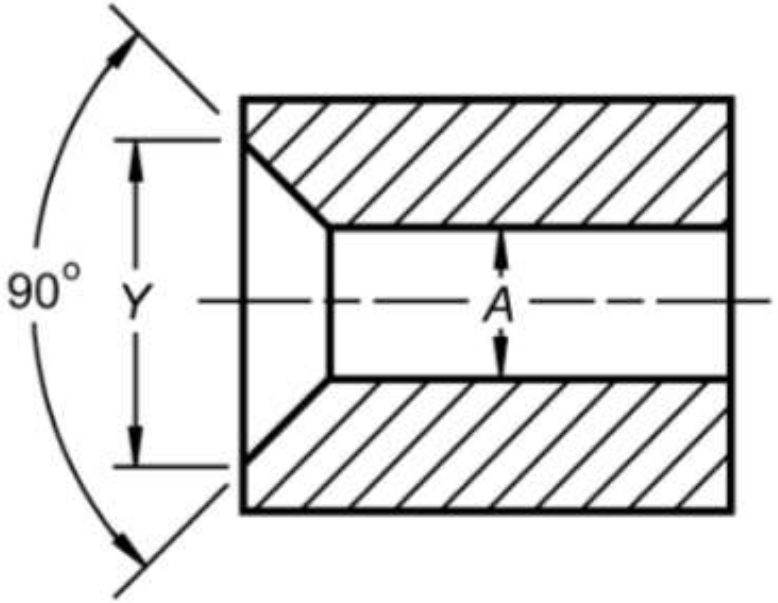
Exercise 10-6

- **M8x1.25 Flat Countersunk Head Metric Cap Screw**

Major dia.	8
Head dia.	A = 17.92
C'Sink dia.	
Close clearance hole dia.	

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Drill and Countersink Sizes for Flat Countersunk Head Cap Screws



D Nominal Screw Size	*A* Nominal Hole Diameter	*Y* Min. Countersink Diameter
M3	3.5	6.72
M4	4.6	8.96
M5	6.0	11.20
M6	7.0	13.44
M8	9.0	17.92
M10	11.5	22.40

Exercise 10-6

- **M8x1.25 Flat Countersunk Head Metric Cap Screw**

Major dia.	8
Head dia.	A = 17.92
C'Sink dia.	**Y = 17.92**
Close clearance hole dia.	

Or, Y = A + 3 = 20

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Exercise 10-6

- **M8x1.25 Flat Countersunk Head Metric Cap Screw**

Major dia.	8
Head dia.	A = 17.92
C'Sink dia.	Y = 17.92
Close clearance hole dia.	

Or, Y = A + 3 = 20

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Metric Clearance Holes

Nominal Screw Size	Fit Classes		
	Normal	Close	Loose
	Nominal Drill Size		
M1.6	1.8	1.7	2
M2	2.4	2.2	2.6
M2.5	2.9	2.7	3.1
M3	3.4	3.2	3.6
M4	4.5	4.3	4.8
M5	5.5	5.3	5.8
M6	6.6	6.4	7
M8	9	8.4	10
M10	11	10.5	12
M12	13.5	13	14.5
M14	15.5	15	16.5
M16	17.5	17	18.5
M20	22	21	24

Exercise 10-6

- **M8x1.25 Flat Countersunk Head Metric Cap Screw**

Major dia.	8
Head dia.	A = 17.92
C'Sink dia.	Y = 17.92
Close clearance hole dia.	**8.4**

Or, Y = A + 3 = 20

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Threads and Fasteners

The End

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